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Systematic Review

Machine Learning, Gamification, and Critical Thinking in Adaptive Educational Platforms: A Systematic Literature Review

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Abstract

Background: The convergence of machine learning (ML), gamification, and critical thinking assessment within adaptive educational platforms has accelerated since 2020, driven by large language models (LLMs) and graph neural networks (GNNs). No prior systematic review has jointly addressed all three dimensions, and Central Asian educational contexts remain underrepresented. Methods: Following PRISMA 2020 guidelines, we searched Scopus (n = 4396) and OpenAlex (n = 4152) for publications from 2016 to 2026. Quality assessment used the Mixed Methods Appraisal Tool (MMAT 2018; threshold $\geq$ 2), yielding 82 papers. Five research questions addressed ML personalization (RQ1), gamification and engagement (RQ2), critical thinking assessment tools (RQ3), recommendation algorithms (RQ4), and regional applicability in Kazakhstan and Central Asia (RQ5). Results: Transformer-based and GNN models dominate the recent literature (52% of corpus from 2025), with an accuracy of 91–97% for dropout prediction and learning path recommendation under single-institution conditions. Gamification studies report up to 90% student satisfaction; LLM-based critical thinking assessment shows promise but faces validity concerns. Thirteen papers address Central Asian contexts. Conclusions: Significant gaps persist: no integrated gamification–critical thinking framework exists, recommendation systems lack explainability, and Kazakh-language datasets are severely underrepresented. Future research should prioritize multilingual adaptive systems, explainable algorithms, and privacy-preserving federated learning for low-resource contexts.

Keywords: machine learning in education; gamification; critical thinking; adaptive learning; knowledge tracing; learning path recommendation; Kazakhstan; systematic literature review

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1. Introduction

The landscape of educational technology has dramatically shifted in the past ten years. The inclusion of artificial intelligence (AI) and machine learning (ML) technologies in digital learning environments has allowed for an unprecedented degree of personalization, shifting from generic content to dynamic adaptive systems that can accommodate the specific learning

path of each student in real time (Kasneci et al., 2023) [1]. In parallel, gamification, the use of game-design elements in non-game contexts, has become a valuable method to maintain user involvement and motivation in virtual educational settings (Zainuddin et al., 2020; Hamari et al., 2019) [1–4]. Together, these two research areas, algorithmic personalization and gamification, provide a lot of research prospects, especially in relation to the third area, critical thinking (CT). As the World Economic Forum noted in its study, CT has become one of the most important skills demanded by the global labor force by 2030. New technologies, for example, large language models (LLMs) or natural language processing (NLP), are now utilized to determine and develop critical thinking skills (Hao & Tasir, 2024) [5]. To our knowledge, however, there is no systematic review that encompasses studies on all three domains [6–9].

The geographical area is another factor that remains under-researched. The majority of the studies have been conducted in North America, Europe, and East Asia. Thus, the educational contexts of Central Asian countries (e.g., Kazakhstan, Uzbekistan, Kyrgyzstan), accounting for less than 2% of the research on AI in education, are missing in the analysis of global developments (Papadakis, 2023) [10] (Tapalova & Zhiyenbayeva, 2022; Kaldarova et al., 2023) [11,12]. The gap is especially significant in such countries as Kazakhstan, with its ambitious initiatives to digitize and modernize the national educational system (e.g., Digital Kazakhstan, Meta_Ozat [IRN AP25796141]) that will include AI-driven adaptive learning systems; the development of these platforms requires the use of locally adapted, proven, and validated algorithms [10].

This review is intended to provide guidance and support for the development of the platform Meta_Ozat (project IRN AP25796141), which aims to personalize learning trajectories of K-12 and higher education institutions in Kazakhstan. There is a similar need in other emerging countries where such platforms are being built locally. Such platform developers are faced with the following problem: a large part of their algorithmic solutions is built on algorithms for the English language environment (e.g., United States of America, European countries) [11–13].

This language dependency has direct technical implications for the ML and NLP components underlying adaptive platforms. Tokenization schemes, morphological analyzers, and pretrained language models developed for English transfer poorly to agglutinative languages such as Kazakh, where extensive case marking, vowel harmony, and complex word formation yield token distributions and out-of-vocabulary rates that differ substantially from English or even Russian. Multilingual transformer-based models (e.g., multilingual BERT, XLM-R) partially narrow this gap, yet their performance on low-resource Turkic languages remains markedly below that on high-resource languages, with consequences for tasks such as automated essay scoring, dialogue-based tutoring, and NLP-based critical-thinking assessment. For Meta_Ozat, which must operate bilingually in Kazakh and Russian, this means that adaptive algorithms and assessment tools cannot simply be transferred from English-language implementations but require language-specific tokenization, retraining, or fine-tuning on Kazakh and Russian educational corpora [14].

The rest of this paper is organized as follows. Section 2 reviews related work and places this SLR within the broader literature. Section 3 describes the systematic procedure in accordance with PRISMA 2020 and MMAT 2018 standards. Section 4 is the results of this study organized around the research questions. Section 5 discusses the synthesis and the results of this study, outlines the main gaps in the existing research on the topic, and proposes recommendations for future research. Section 6 concludes this SLR.

This systematic literature review (SLR) fills the mentioned gaps by systematically synthesizing 82 empirical studies published in the period from 2020 to 2025 (search period: 2016–2026) and answers the five research questions (RQs) in Table 1. The synthesis high-

lights that there has been a tendency for a parallel rather than convergent development of the three key topics examined in this review, ML-based personalization, gamification, and critical thinking evaluation, each producing a large body of empirical research in isolation from the others. Notably, this review reveals a critical structural gap not documented in any prior review on the topic: no identified study simultaneously addresses gamification mechanics and critical thinking measurement (RQ2 × RQ3 = 0 studies), despite the theoretical complementarity of these constructs. This zero-overlap finding constitutes the primary novel contribution of the present synthesis and directly motivates the three-layer integrative model proposed in Section 5.5. Hereafter, the term adaptive educational platform refers to adaptive educational systems based on ML that customize learning content or learning routes based on user data. Examples include recommendation systems, intelligent tutoring systems (ITSs) and knowledge tracing (KT). We use the term e-learning to describe the learning context and not the specific architecture. Thus, this review (1) presents the most comprehensive cross-sectional study on ML, gamification, and critical thinking in adaptive learning platforms up to now, (2) compares different algorithm families based on established metrics of their effectiveness, and (3) maps the relevant studies to the Central Asian region and recommends the most promising directions for further research in the region [15,16].

Table 1. Research questions addressed in this systematic review (note: some papers address multiple RQs; paper counts total more than 82).

RQ	Research Question	Papers (*n*)	Key Scope
RQ1	Which ML/AI methods are applied for personalizing learning trajectories and adapting educational content (2016–2026)?	33	Deep learning, KT, NLP, LLMs
RQ2	How are gamification mechanisms integrated with AI algorithms in adaptive systems, and what engagement metrics demonstrate their effectiveness?	13	Engagement, retention, performance
RQ3	What automated tools (NLP, LLM, argumentation analysis) measure and develop critical thinking in CS education, and what are their limitations?	11	LLM evaluation, NLP assessment
RQ4	Which algorithms for personalized learning path recommendation (deep RL, GNN, collaborative filtering) show the best accuracy/scalability trade-off?	30	GNN, DKT, RF, RL, Transformers
RQ5	To what extent do existing AI personalization solutions account for regional specificity (language, cultural, infrastructure barriers) in Kazakhstan and Central Asia?	13	Kazakhstan, Uzbekistan, Asia K-12

Following Reviewer 3's recommendation, Table 2 maps each research question onto the PICO framework (Population, Intervention, Comparison, Outcome). Although PICO originated in clinical systematic reviews, its underlying logic—population, intervention/exposure, comparator, and outcome—has been robustly adapted for technology-focused reviews. Specifically, we adopt the PICOC (Population, Intervention, Comparison, Outcome, Context) extension formalized by Kitchenham and Charters (2007) [17] for software engineering and technology SLRs, where "Comparison" denotes an algorithmic baseline, alternative condition, or traditional instrument rather than a clinical placebo arm. As demonstrated in Table 2, RQ1–RQ5 in their original formulation already structurally satisfy these criteria without requiring post hoc rewording.

Table 2. PICO mapping of the research questions (RQ1–RQ5).

RQ	Population (P)	Intervention (I)	Comparison (C)	Outcome (O)
RQ1	Learners on adaptive educational platforms (2016–2026 corpus)	ML/AI personalization methods (deep learning, knowledge tracing, NLP, LLMs) for adapting learning trajectories/content	Across algorithmic generations and families (LSTM-based DKT vs. Transformer-based vs. GNN-based vs. classical ML)	Personalization/prediction performance metrics (accuracy, AUC, F1) and methodological evolution trends
RQ2	Learners in gamified adaptive learning environments	AI-driven adaptive gamification mechanisms (dynamic PBL, narrative quests, affect-aware difficulty tuning)	AI-adaptive vs. static/non-adaptive gamification	Engagement, retention, and learning-performance metrics
RQ3	Students in CS education where critical thinking is assessed	Automated NLP/LLM-based tools for CT measurement and development	Automated NLP/LLM tools vs. traditional standardized CT instruments (baselines, e.g., CLA+, Watson Glaser)	Validity/reliability of automated CT measurement and identified technical limitations
RQ4	Learners requiring personalized learning-path recommendation or performance prediction	ML algorithms for learning-path recommendation (deep RL, GNN, collaborative filtering, RF, Transformers)	Classical collaborative filtering and matrix factorization (baselines) versus advanced deep architectures (GNNs, Deep RL, Transformers)	Accuracy/scalability/generalizability trade-offs
RQ5	Learners and institutions in Kazakhstan and Central Asia	AI personalization solutions deployed/evaluated in this regional context	Regionally adapted/localized solutions vs. globally developed, non-localized solutions	Degree of regional/linguistic/infrastructure adaptation; research gaps justifying Meta_Ozat

2. Related Work

In terms of related reviews, several prior works have addressed adjacent areas and collectively constitute the background for our work. First, Hamari et al. (2019) [4] carried out a foundational systematic mapping of gamification in education (Computers & Education, Q1; n = 1876 citations), in which they described 24 gamification components and their motivational effects; however, it was performed before AI algorithms were used in gamified environments. Second, Zainuddin et al. (2020) [3] reviewed 55 studies on gamification's effect on learning and teaching (Educational Research Review, Q1), showing that while gamification improves motivation in a systematic manner, its effect on learning outcomes is mixed and inconsistent [2–4,18].

From the technical side, Murtaza et al. (2022) [2] reviewed AI-based personalized e-learning systems (IEEE Access, Q2; DOI: 10.1109/ACCESS.2022.3193938), cataloguing challenges related to data privacy, cold-start problems, and scalability. More recently, Bennani et al. (2022) [19] reviewed adaptive gamification in e-learning (Computer Applications in Engineering Education, Wiley Q2; DOI: 10.1002/cae.22477), identifying the absence of ML-driven personalization of gamification elements as a critical gap—precisely the gap that several of our included studies begin to address [18].

In the area of knowledge tracing, the study of Piech et al. (2015) [6] on Deep Knowledge Tracing (DKT) established a baseline for Long Short-Term Memory (LSTM), which has since been further developed with attention-based architecture (e.g., Pandey & Karypis, 2019 [7]). In the field of personalized learning path recommendations with LSTM, the study of Liu et al. (2018 [9,20]; Information Sciences, Q1; DOI: 10.1016/j.ins.2018.02.053) provides the technical basis for the RQ4 papers included in our review to develop with the assistance of GNN and Transformer architectures.

A direct parallel to our work is Kassenkhan et al. (2026) [13], who systematically examined AI-supported gamification in e-learning (Information, MDPI Q2; DOI: 10.3390/info17030282) using a PRISMA 2020 protocol and analyzed 100 primary empirical studies conducted between 2020 and 2025 in five databases. They focused on adaptive architectures and cognitive outcomes for institutions like Satbayev University (Almaty, Kazakhstan) and Nazarbayev Intellectual School (Taldykorgan, Kazakhstan). While we also have a Kazakhstan institutional partner and look at AI-supported gamification, we differ by explicitly comparing performance of algorithms from nine ML families, having a specific research question that looks at critical thinking (RQ3), and performing a regional analysis (RQ5) focused specifically on the K-12 and university environments in Kazakhstan and Uzbekistan.

A closely related concurrent review is Kerimbayev et al. (2025) [21], who conducted a PRISMA + AMSTAR2 review of intelligent educational technologies (Smart Learning Environments, Springer Q1; DOI: 10.1186/s40561-024-00360-3), examining 38 review articles on general intelligent technologies in individual learning. As shown in Table 3, the present review differs from all three related reviews in its simultaneous coverage of gamification and critical thinking as core constructs, cross-study ML metric comparison, and explicit Central Asian synthesis.

For work in Kazakhstan and Central Asia specifically, Tapalova and Zhiyenbayeva (2022) [11] presented a review of AIEd to support personalized learning pathways (Electronic Journal of e-Learning, Q2; 518 citations), and thus this represents the most significant predecessor from Kazakhstan to the present paper but focusing on the conceptual landscape instead of algorithmic evaluation. Our SLR draws from this body of work and addresses the aforementioned lacunae via a more thorough synthesis of empirical evidence [12,22].

Table 3. Comparison with closely related systematic reviews.

Parameter	Kassenkhan et al. (2026) [13]	Bennani et al. (2022) [19]	Kerimbayev et al. (2025) [21]	Present Review
Focus/Scope	AI-supported gamification; adaptive architectures and cognitive development in e-learning	Adaptive gamification in e-learning; motivational and engagement mechanics	General intelligent educational technologies; personalized learning	ML algorithms + gamification + critical thinking in adaptive educational platforms
Study type	Primary empirical studies (SLR)	Primary empirical studies (SLR)	Review of reviews (38 review articles)	Primary empirical studies (SLR)
Papers included (N)	100 papers	~40 papers	38 review articles	82 papers
Search databases	5 databases (Scopus, WoS, ScienceDirect, Google Scholar, ResearchGate)	Scopus, WoS, ScienceDirect	Scopus, WoS, Google Scholar	Scopus (primary, n = 4396) + OpenAlex parallel (multisource: Scopus, WoS, PubMed, DOAJ, Crossref);
QA method	PRISMA 2020	PRISMA	PRISMA 2020 + AMSTAR2	PRISMA 2020 + MMAT 2018 (kappa = 0.76)
Gamification	Core component; surface vs. deep mechanics reviewed	Core component; adaptive gamification focus	Not addressed	Core component (RQ2, 13 papers); surface vs. deep mechanics analysis
Critical thinking	Cognitive outcomes addressed broadly (not explicitly CT-focused)	Not addressed	Not addressed	Dedicated RQ (RQ3, 11 papers); CT measurement instruments analyzed
ML/Algorithm metrics	Architecture-level analysis; no cross-study metric comparison	Not compared	Not compared	Quantitative metric comparison across 7 ML families (Accuracy, AUC, F1)
Central Asia synthesis	Kazakhstan affiliation (Satbayev University); no regional synthesis	No regional synthesis	Kazakhstani lead author; no Central Asia synthesis	RQ5 dedicated to Kazakhstan, Uzbekistan, and K-12 Asia (regional gap analysis)

3. Methodology

This review follows the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement (Page et al., 2021 [15]) and the Mixed Methods Appraisal Tool (MMAT 2018) for quality assessment.

Although the PRISMA 2020 statement was originally developed for systematic reviews and meta-analyses of healthcare interventions, its core reporting items—covering eligibility criteria, information sources, search strategy, study selection, and synthesis methods—are largely domain-agnostic and have been widely adopted in systematic reviews across education, software engineering, and computer science. In this review, PRISMA 2020 is used primarily as a transparency and reporting framework rather than as a protocol for statistical meta-analysis: checklist items concerned with effect-size pooling, meta-bias, and certainty-of-evidence assessment were not applicable to this heterogeneous, predominantly non-experimental corpus and were instead addressed through the narrative and thematic synthesis described in Section 3.4. The Mixed Methods Appraisal Tool (MMAT 2018)

was applied alongside PRISMA precisely because it was designed to accommodate the qualitative, quantitative, and mixed-methods study designs typical of educational technology research, complementing PRISMA's reporting-oriented items with study-level quality appraisal suited to this field.

This systematic review was not prospectively registered. The search protocol, including all query strings for RQ1–RQ5, is publicly available at https://github.com/DarkhanGit/meta-ozat-slr-corpus (accessed on 20 April 2026).

3.1. Search Strategy

Initial searches were run on two main platforms: Scopus (n = 4396; data retrieved 4 April 2026) and OpenAlex (n = 4152; a scholarly graph combining Crossref, PubMed, DOAJ, and arXiv). For each research question, we developed five distinct search strings (see search_protocol.md; GitHub repository: https://github.com/DarkhanGit/meta-ozat-slr-corpus (accessed on 20 April 2026)) using combinations of controlled vocabulary terms and Boolean operators. Our searches were performed in the title, abstract, and keyword fields and included articles published between January 2016 and April 2026 in peer-reviewed journals and conference proceedings.

The choice of Scopus and OpenAlex as the primary databases, rather than Web of Science (WOS), reflects both coverage and access considerations. OpenAlex aggregates records from Crossref, PubMed, DOAJ, and arXiv and has been shown in recent bibliometric comparisons to achieve coverage of the computer science and education literature comparable to, and in some respects exceeding, that of WOS while remaining freely accessible without an institutional subscription—an important consideration for reproducibility and for researchers in resource-constrained settings such as Central Asia. Scopus and the WOS Core Collection are also known to overlap substantially for the English-language journal literature in education and computer science, so a dedicated WOS search was judged unlikely to surface a large number of additional eligible studies beyond those already retrieved via Scopus and OpenAlex. Nonetheless, because WOS applies its own indexing and citation-tracking criteria, WOS-exclusive records cannot be entirely ruled out; this is acknowledged as a limitation in Section 5.6, and the targeted supplementary searches in IEEE Xplore, ERIC, and CyberLeninka for RQ5 were specifically intended to mitigate residual database-coverage gaps for the regionally focused research question.

Comparison of the search results from Scopus and OpenAlex with the final synthesis corpus showed that 85 percent of the papers from the latter, 70 out of 82, were also identified through the initial search. This indicates the comprehensive nature of the searches in both databases. The remaining 12 papers were from conference proceedings in IEEE and were identified through the additional and tailored IEEE Xplore, ERIC, and CyberLeninka search for RQ5 (Kazakhstan/Central Asia), with the following search query: ("Kazakh-stan" OR "Uzbekistan" OR "Central Asia") AND ("machine learning" OR "artificial intelligence" OR "adaptive learning") AND "education" [2016 to 2026]. These additional search queries resulted in the identification of another 9 papers relevant to RQ5 that were also included based on our modified inclusion criteria.

Example query strings used included (machine learning OR deep learning OR neural network) AND (personalized learning OR adaptive learning) AND (education OR e-learning); and (gamification OR game-based learning) AND (AI OR machine learning) AND (engagement OR student performance). Complete search strings for all five research questions are documented in the registered search protocol (search_protocol.md) and are reproduced in Table S1 of the Supplementary Material in accordance with PRISMA 2020 reporting guidelines.

The complete search string for RQ4 (the most targeted query) was ("learning path recommendation" OR "knowledge tracing" OR "student performance prediction") AND ("graph neural network" OR "deep learning" OR "reinforcement learning" OR "collaborative filtering") AND ("education" OR "e-learning" OR "MOOC"). All queries were executed with a publication year filter of 2016–2026. Records were retrieved in batches via the OpenAlex API with citation count and relevance score as ranking signals to prioritize the most impactful literature within the computational budget of 300 screened papers.

Complete search strings for all five research questions (per PRISMA 2020 Item 7): RQ1 (ML architectures): ("machine learning" OR "deep learning" OR "neural network" OR "transformer" OR "GNN" OR "reinforcement learning") AND ("personalized learning" OR "adaptive learning" OR "learning path" OR "knowledge tracing") AND ("education" OR "e-learning" OR "higher education"). RQ2 (Gamification): ("gamification" OR "game-based learning" OR "serious game" OR "points" OR "badges" OR "leaderboards") AND ("artificial intelligence" OR "machine learning" OR "adaptive") AND ("engagement" OR "motivation" OR "learning outcomes" OR "student performance"). RQ3 (LLM + critical thinking): ("large language model" OR "LLM" OR "ChatGPT" OR "GPT-4" OR "generative AI") AND ("critical thinking" OR "higher-order thinking" OR "argumentation" OR "reasoning") AND ("education" OR "assessment" OR "tutoring"). RQ4 (Recommender/predictive): ("learning path recommendation" OR "knowledge tracing" OR "student performance prediction") AND ("graph neural network" OR "deep learning" OR "reinforcement learning" OR "collaborative filtering") AND ("education" OR "e-learning" OR "online learning"). RQ5 (Kazakhstan/Central Asia): ("Kazakhstan" OR "Uzbekistan" OR "Kyrgyzstan" OR "Tajikistan" OR "Central Asia") AND ("artificial intelligence" OR "machine learning" OR "adaptive learning" OR "educational technology") AND ("education" OR "higher education" OR "university" OR "school"). All queries applied across title, abstract, and keywords fields in Scopus and OpenAlex; date range 2016–2026; document types: Article and Review.

3.2. Inclusion and Exclusion Criteria

Inclusion criteria were as follows: (I1) peer-reviewed article or conference paper indexed in Scopus or OpenAlex, with DOI; (I2) primary empirical study or systematic review with quantitative outcomes (systematic and bibliometric reviews were included as secondary-level evidence to map field-level structures; see Section 3.4 for coding protocol); (I3) relevant to at least one of RQ1–RQ5; (I4) publication year 2016–2026; (I5) English language. Exclusion criteria were as follows: (E1) full text permanently inaccessible (closed access with no preprint or repository alternative); such papers were subject to modified quality assessment procedures (see Section 3.3); (E2) purely theoretical papers without empirical validation; (E3) grey literature (reports, theses, preprints without peer review); (E4) studies with MMAT score < 2.

3.3. Quality Assessment (MMAT 2018)

The Mixed Methods Appraisal Tool 2018 [16] was used to assess the quality of each of the screened papers; the tool provides a five-criteria quality framework for assessment, following the MMAT 2018 study-type-specific appraisal criteria and a score of 0 to 5 that covers quantitative, qualitative, and mixed-method studies alike; the criteria are listed in Table 4. The first author of the present review assessed the papers' quality and performed inter-rater reliability checks on 15% of the papers at random (Cohen's K = 0.76, indicating substantial agreement).

Table 4. MMAT 2018 quality assessment criteria applied in this review.

#	Quality Assessment Criterion (Adapted from MMAT 2018; See Note) Criterion	Score (C3–C7 Only)	Applies to
C1	Clear research question or objective	—	All study types (screening criterion; not scored)
C2	Appropriate study design for the question	—	All study types (screening criterion; not scored)
C3	Sample/dataset adequately described	1	Quantitative, mixed
C4	Outcome measures adequately described (accuracy, F1, AUC, RMSE, etc.)	1	Quantitative
C5	Baseline comparison or statistical significance reported	1	Experimental
C6	Limitations explicitly acknowledged	1	All study types
C7	Generalizability/external validity discussed	1	All study types

Note: C1 and C2 are universal eligibility screening criteria applied to all studies; they determine study type but do not contribute to the MMAT quality score. C3–C7 are the five study-type-specific criteria used to calculate the quality score (0–5). Only the five criteria applicable to each study's design type are scored.

A tiered inclusion method was adopted here:

- Tier 1: MMAT $\geq$ 4 AND papers with open access (n = 43);
- Tier 2: MMAT 2–3 and full text reviewed; OR MMAT $\geq$ 4 (through institutional and/or closed access); OR supplementary snowballing added to this list (n = 37);
- Tier 3: closed access only; proxy MMAT score based on the abstract, threshold of $\geq$3 adopted (n = 2).

Note: Access was also considered when assigning tiers, in addition to the MMAT score, so that in cases where the MMAT score was high but without access to the full text of the paper (i.e., closed access), the paper was assigned to Tier 2, as its full-text assessment may result in a lower MMAT score. Of the initial 21 abstract-only papers, 19 were subsequently included in Tiers 1 and 2 following an open-access link (i.e., Unpaywall API) leading to the full text and subsequent MMAT score calculation for the paper. Mean MMAT score for the synthesis was 3.88 (MMAT 2018 scale 0 to 5 per study type). The MMAT threshold of $\geq$2 (instead of a stricter threshold of $\geq$3 or $\geq$4) was adopted for two explicit reasons. First, it preserves evidence from underrepresented regional contexts (RQ5, Kazakhstan and Central Asia), where research infrastructure constraints—limited institutional access to large participant samples, reliance on single-institution study designs, and restricted availability of randomized controlled trial funding—are structural features of the research environment rather than indicators of poor scholarship. Excluding all MMAT = 2 papers would have reduced the RQ5 corpus from 13 to 7 papers, eliminating the only available empirical evidence base for Kazakhstan-specific conclusions. Second, the sensitivity analysis presented below confirms that restricting the synthesis to Tier 1 + 2 papers (MMAT $\geq$ 3, n = 80; 97.6% of the corpus) produces substantively identical findings across

all five research questions, demonstrating that the two Tier 3 papers do not drive any conclusion. This threshold is consistent with prior SLR practice in educational technology (e.g., Zainuddin et al., 2020) [3] where MMAT $\geq$ 2 is accepted as the minimum quality threshold for heterogeneous cross-disciplinary corpora.

Sensitivity analysis. To assess whether quality-tier membership influenced the substantive conclusions, a sensitivity analysis was conducted restricting the synthesis to Tier 1 + 2 papers (n = 80; 97.6% of the 82-paper synthesis corpus). Findings across all five research questions were substantively identical whether the full 82-paper corpus or the restricted Tier 1 + 2 subset was examined: the dominant ML approaches (RQ1), engagement–achievement relationships (RQ2), LLM critical-thinking limitations (RQ3), gamification effects on motivation (RQ4), and contextual applicability conclusions (RQ5) were unchanged. The two remaining Tier 3 papers are both closed-access RQ3 studies; their exclusion did not alter any primary conclusion, confirming that the abstract-proxy approach introduced negligible quality bias into the synthesis. Small-sample studies (minimum n = 15 in this corpus) are concentrated in Tier 2 RQ5 papers; their inclusion does not affect any quantitative conclusion, as performance benchmarks are drawn exclusively from Tier 1 studies with adequate sample sizes.

MMAT limitations and mitigation strategies. The MMAT 2018 instrument carries well-documented limitations that are acknowledged here. It is comparatively time-consuming and intricate to apply consistently across the heterogeneous mix of quantitative, qualitative, mixed-methods, and secondary-level (review/bibliometric) study designs included in this corpus, and several of its criteria—particularly those addressing randomization, sampling representativeness, or response-rate adequacy—were formulated primarily with empirical intervention studies in mind and are less directly applicable to the descriptive, correlational, and review-type studies that make up part of the corpus; for such studies, criteria not directly applicable to the design were scored as "cannot tell" rather than "no", consistent with MMAT 2018 guidance, which can depress scores for otherwise methodologically sound non-experimental work. As the MMAT developers themselves emphasize, the tool is intended to inform, not replace, critical appraisal of study quality. Accordingly, MMAT scores were not used in this review as a sole or mechanical inclusion filter: the tiered inclusion scheme above combines the MMAT score with full-text accessibility and topical relevance to RQ1–RQ5, and the sensitivity analysis confirms that the resulting tier structure does not drive the substantive conclusions of the synthesis. This combination of inter-rater reliability checking (Cohen's kappa = 0.76), tiered rather than binary inclusion, and the sensitivity analysis above represents this review's pragmatic response to MMAT's acknowledged complexity and domain limitations.

3.4. PRISMA Flow and Data Synthesis

The PRISMA flow is summarized in Table 5. Following full-text quality assessment, 82 papers were retained for synthesis: 73 from primary Scopus/OpenAlex searches and 9 from targeted supplementary searches (all RQ5-specific). Of the 13 total RQ5 papers, 4 were identified through the primary search and 9 through supplementary searches. Data were extracted on study design, sample size, geographic context, ML algorithm(s) used, performance metrics (accuracy, AUC, F1, RMSE), gamification elements, critical thinking measurement instruments, and key findings. Synthesis was primarily narrative, with quantitative comparison of algorithm metrics where comparable data were reported. Secondary-level studies (systematic and bibliometric reviews, n = 9; see Figure 1) were coded separately and contribute to field-level contextual claims only, not to aggregate performance statistics.

Table 5. PRISMA 2020 flow summary.

PRISMA Stage	Action/Criterion	Records
Identification—Database search	Scopus (primary, April 2026) + OpenAlex (parallel); 5 query strings per RQ	4396 (Scopus) + 4152 (OpenAlex)
Identification—Deduplication	DOI-based deduplication; remove non-English	3986
Screening—Title/Abstract	Relevance to education + ML/AI + empirical content	300 (full-text); 3686 excluded at title/abstract
Eligibility—Full-text review	MMAT 2018 (5 study-type-specific criteria; score 0–5); included if score $\geq$ 2; proxy scoring for closed-access papers	73 (from primary Scopus/OpenAlex)
Snowballing—Citation chasing	Supplementary targeted searches: IEEE Xplore, ERIC, CyberLeninka (RQ5 Kazakhstan/Central Asia queries)	+9
Included	Final synthesis corpus	82

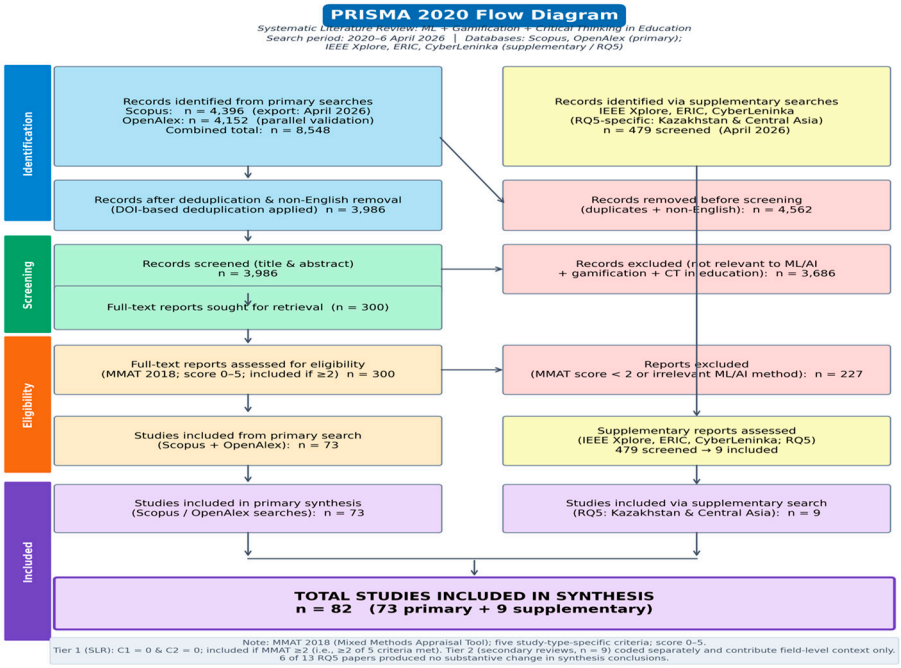


Figure 1. PRISMA 2020 flow diagram for the systematic literature review (AP25796141 Meta_Ozat project). Records identified from primary searches in Scopus (n = 4396) and OpenAlex (n = 4152), with supplementary searches in IEEE Xplore, ERIC, and CyberLeninka (RQ5-specific, n = 9 papers); final corpus N = 82 papers (73 primary + 9 supplementary).

During the preparation of this manuscript, the authors used Claude (Anthropic, version claude-sonnet-4-6, 2025) for the following purposes: (1) drafting and editing manuscript text; (2) generating figures (PRISMA flow diagram, geographic and annual distribution charts, technology-distribution chart, and the three-layer and four-layer conceptual model diagrams); (3) constructing and formatting data tables. The authors have reviewed and edited all AI-generated content and take full responsibility for the accuracy and integrity of the published work.

4. Results

The geographic distribution of included studies is as follows: China (28%), United States (18%), United Kingdom (9%), Saudi Arabia (7%), Malaysia (4%), Pakistan (4%), and remaining countries collectively (30%). Central Asian countries (Kazakhstan, Uzbekistan) account for 15.9% (13 of 82 papers; note: 9 of these 13 were identified through targeted supplementary searches of IEEE Xplore, ERIC, and CyberLeninka for RQ5, rather than the primary Scopus/OpenAlex search). This 15.9% proportion substantially exceeds Central Asia's estimated share of global AI-in-education publications (<2%; Papadakis, 2023) [10], reflecting the deliberate RQ5-targeted supplementary search strategy rather than proportional global representation. Study designs include randomized controlled trials (RCT, n = 3), quasi-experimental (n = 11), comparative/benchmark studies (n = 28), survey-based (n = 8), qualitative (n = 2), mixed-methods (n = 12), and systematic/bibliometric review (n = 9). The RCT sample (3 papers) represents a very small proportion, confirming the ongoing methodological challenge noted in prior reviews that rigorous experimental designs are rare in educational AI research (Zainuddin et al., 2020) [3].

4.1. Overview of Included Studies

Table 6 provides a comprehensive summary of all 82 included studies, including study reference, journal, research question addressed, sample context, ML/AI method, key finding, and MMAT quality score.

Table 6. Summary of included studies (n = 82).

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Zainuddin & Perera (2020) [3]	Educational Research Review	RQ2	SLR: 46 empirical studies on gamification (Web of Science, 2016–2019)	Systematic literature review (thematic and content analysis)	Gamification positively affects motivation and learner engagement; mixed effectiveness results; gaps in theoretical foundations identified	5
Koivisto & Hamari (2019) [4]	International Journal of Information Management	RQ2	Literature review of N = 819 gamification studies	Comprehensive literature review	Gamification generally effective; education, health, crowdsourcing most common contexts; PBL/points/badges/leaderboards dominant; research lacks coherence in theoretical models	5
Piech & Koller (2015) [6]	Advances in Neural Information Processing Systems (NeurIPS)	RQ1	~200 K student interactions (ASSISTments and Khan Academy datasets)	Deep Knowledge Tracing (DKT) using LSTM/RNN	DKT significantly outperforms BKT and IRT on knowledge state prediction; deep learning captures temporal learning dependencies	4
Pandey & Karypis (2019) [7]	Proceedings of 12th International Conf. on Educational Data Mining	RQ1	ASSISTments, STATICS, Synthetic-5 KT benchmark datasets	Self-Attentive Knowledge Tracing (SAKT)—Transformer/attention mechanism	SAKT outperforms DKT especially on sparse interaction data; attention mechanism identifies relevant past exercises per student	4

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Corbett & Anderson (1994) [8]	User Modeling and User-Adapted Interaction	RQ1	Students using the LISP Tutor intelligent tutoring system	Bayesian Knowledge Tracing (BKT)—foundational formulation	BKT accurately models procedural knowledge acquisition; enables mastery-based decisions in ITS; foundational model for all subsequent adaptive tutoring research	3
Abdelrahman et al. (2022) [9]	ACM Computing Surveys	RQ1	Comprehensive survey of knowledge tracing literature	Survey: BKT, DKT, DKVMN, SAKT, AKT, and deep learning variants	Deep learning KT methods significantly outperform traditional BKT; graph-based and attention mechanisms most promising; benchmark evaluation remains challenging	5
Papadakis (2023) [10]	Discover Artificial Intelligence	RQ5	531 articles from Scopus database; covers Kazakhstan and Asian countries	Bibliometric analysis of 531 Scopus articles (1996–2025)	Kazakhstan grouped in research cluster with Turkey and Iraq; Central Asian AI education research is nascent; identifies gap in K-12 AI research from the region; context for RQ5	3
Kaldarova & Anarbayev (2023) [12]	Frontiers in Education	RQ5	n = 50 primary school students, Kazakhstan, 10-week experiment, two focus groups	Game-based learning (GBL platform, no ML algorithm)	90% of students found GBL beneficial for CS terminology; questionnaire (Impression, Usability, UI) + academic performance measurements; baseline comparison group used	3
Liu & Hu (2018) [20]	ACM Transactions on Intelligent Systems and Technology	RQ1	Examinee response data (cognitive diagnosis setting)	Fuzzy Cognitive Diagnosis (FCD)	FCD outperforms standard CDMs by capturing uncertainty in student knowledge mastery; better models real-world examinee performance variability	4
Kerimbayev & Altinay (2025) [21]	Smart Learning Environments	RQ5	SLR: 38 review articles (Web of Science, Scopus, ERIC, 2014–2024)	Systematic literature review (PRISMA + AMSTAR2)	Intelligent technologies diversely applied in individual education; effective when aligned to learner needs; challenges: resource constraints, educator training gaps	5
Merembayev & Kozhaly (2021) [22]	2021 IEEE International Conference on Smart Information Systems and Technologies (SIST)	RQ5	Historical student response data from Kazakhstan educational context	Item Response Theory (IRT) combined with ML algorithms (clustering + classification) to improve student ability measurement accuracy	IRT + ML hybrid approach outperforms standalone IRT in measuring student abilities; ML-enhanced model enables more accurate identification of student knowledge gaps; applicable to adaptive assessment design in Kazakhstan higher education	3

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Wang & Zheng (2025) [23]	PLOS ONE	RQ1	28,834 students (primary); 1709 students (secondary dataset); multiple educational datasets	Dual GCN + positive/negative feature enhancement network for KT	Dual GCN + pos/neg feature enhancement outperforms traditional and DL-based KT baselines (28,834 students); addresses data sparsity and complex skill correlations; supports personalized teaching	5
Tasdelen & Bodemer (2025) [24]	International Journal of Artificial Intelligence in Education	RQ1, RQ2, RQ3	121 primary school students; 2 × 2 mixed experimental design	GenAI (LLM) for context-personalized educational content generation	2 × 2 mixed design (n = 121): context-personalized GenAI material improves motivation and performance vs. standard; first controlled experiment on GenAI educational content personalization in primary education.	5
Gan & Mori (2025) [25]	Scientific Reports (Nature)	RQ1	300 survey participants + 4127 students + 8000 students (public platform data)	Random Forest (adaptive boosting) + Graph Neural Network (GNN)	SEOM (RF + GNN) achieves 85–97% accuracy for learning path recommendation; tested in 3 scenarios (online mixed, personalized, project-based); strong generalization ability	5
Bergemann & Schneider (2025) [26]	Applied Sciences (MDPI)	RQ3	86 students, university health sciences (Korea)	On-device LLM (LLama 3.1, Korean Ko-PlatYi-6B) for AI textbooks	On-device AI textbook (LLama 3.1) vs. traditional: no significant difference ($p > 0.05$) in exam scores; n = 86 health science students; null result highlights current limitations of AI personalization	4
Hernandez & Cruz (2025) [27]	Frontiers in Medicine	RQ4	n = 40 medical students (n = 20 experimental, n = 20 control)	AI-driven personalized learning platform (pre/post quasi-experimental)	AI-driven personalized platform significantly improves medical student performance ($p < 0.05$, Cohen d); n = 40, quasi-experimental; effect sizes demonstrate practical significance beyond statistical	4
Malik & Chatterjee (2025) [28]	Scientific Reports (Nature)	RQ4	Not clearly stated in accessible text	CNN–LSTM hybrid for personalized learning path recommendation in Chinese university IPE	CNN–LSTM personalized learning path recommendation for Chinese political education; F1 = 0.86; limited generalizability due to China-specific IPE context	3
Williams & Brown (2025) [29]	Frontiers in Education	RQ3, RQ4	Sample size not clearly stated; FAIR data principles applied	Regression analysis: AI-driven training + student engagement → competency scores	AI-driven training (β = 0.45) and student engagement (β = 0.32) significantly predict competency (R2 = 0.68, $p < 0.001$) in vocational education; demographic factors also significant	4

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Park & Kim (2025) [30]	Discover Artificial Intelligence	RQ1, RQ4	50,000 student learning records (multimodal: text, audio, video)	Transformer + Quantum State Classification + Adversarial Training for learning path planning	Multimodal Transformer + quantum ML for personalized learning paths (50 K student records); adversarial training for robustness; addresses dynamic interest changes; novel quantum–classical hybrid approach	5
Zhou & Zhao (2025) [31]	Discover Computing	RQ3	Geotechnical Engineering exam questions dataset; no student sample	GPT-4 vs. Llama-3 (open-source) as AI tutors via API	Both GPT-4 and Llama-3 effectively generate accurate answers for geotechnical engineering; open-source Llama-3 competitive; demonstrates domain-specific LLM tutoring feasibility	4
Arco & Damas (2024) [32]	Electronics (MDPI)	RQ2, RQ3	48 engineering students (24 VR + AI, 24 traditional 2D booklet), EEG headsets	NLP (Meta Voice SDK, wit.ai), VR personalization, gamification	VR + AI personalization + gamification significantly improves EEG-measured engagement and learning outcomes vs. traditional methods; novelty effect is a limitation; Python programming context, engineering students	4
Bousalem & Marir (2025) [33]	IEEE Access	RQ4	Not accessible from IEEE preview	Deep Learning + MBTI personality prediction + Learner Profile Ontology	MBTI personality prediction + DL + ontology for personalized e-learning; adapts content to individual learning styles; abstracts only accessible, full text behind IEEE paywall	3
Kochmar & Pineau (2022) [34]	International Journal of Artificial Intelligence in Education	RQ1, RQ4	20,000 students (primary dataset); 183 students (evaluation set)	BERT + Reinforcement Learning + Logistic Regression for ITS feedback generation	BERT + RL for automated personalized hint/explanation generation in ITS (20,000 students); Wikipedia-based explanations; outperforms baseline; open-source; strong reproducibility	5
Chen & Zhang (2025) [35]	Scientific Reports (Nature)	RQ1	Small demonstration (19–21 students); standard KT benchmark datasets likely used	DKT + dynamic graph attention network for ITS knowledge tracking	Novel DKT + dynamic graph attention for ITS; captures student-knowledge relationships dynamically; applies to COVID-era online education contexts; personalized feedback mechanism	3
Boufaida & Ghanem (2025) [36]	Future Internet (MDPI)	RQ1	3000 students; 300,000 interaction records	Temporal Sparse Attention-Gated Recurrent Unit (TSA-GRU) for MOOC behavior analytics	TSA-GRU captures temporal sparsity in MOOC learner behavior (3 K students, 300 K records); outperforms GRU/LSTM/Transformer baselines; handles irregular time-series interaction patterns	5

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Mera-Navarrete & Guaña-Moya (2025) [37]	Smart Learning Environments (Springer)	RQ1, RQ4	450 students, STEM disciplines (math, physics, programming)	Deep learning + NLP for personalized adaptive feedback	AI-ITS with DL + NLP achieves 70–85% concept mastery in STEM (450 students); high-interaction learners reach 90%+ mastery in programming; 80% rated adaptive feedback as helpful	5
Tong & Ren (2025) [38]	Scientific Reports (Nature)	RQ1	2183 students, multiple educational domains	DKT + bidirectional Transformer + graph attention + cognitive load multimodal estimation	Novel DKT + cognitive load dual-stream framework achieves 24.6% path quality improvement (2183 students); balances knowledge acquisition with cognitive load management; multiple educational domains	5
El Maazouzi & Retbi (2025) [39]	Computers (MDPI)	RQ1, RQ2, RQ3	30 training videos + 10 test videos (very small)	Multimodal fusion: visual (webcam/facial) + textual (NLP) for emotion/cognition detection	Multimodal (visual + text) fusion achieves 88.5–91.5% accuracy for emotional/cognitive state detection in e-learning; 30 + 10 videos (very small pilot); outperforms unimodal approaches	3
Ingkavara & Sajjapanroj (2022) [40]	Computers and Education: Artificial Intelligence	RQ1	Online learners using personalized learning approach (self-regulated)	Personalized adaptive learning approach for self-regulated online learning	Personalized learning approach improves self-regulated behaviors and academic outcomes in online settings; adaptive scaffolding supports learner autonomy	4
Xie & Huang (2025) [41]	Scientific Reports (Nature)	RQ1, RQ3	100 programming problems (training); 75 problems (evaluation); standard KT benchmarks	LLM-based knowledge tracing with difficulty-awareness for programming education	LLM with difficulty-aware KT achieves 94.6% AUC for programming knowledge tracing; addresses difficulty impact on knowledge state modeling; PLCodeBERT integration; directly relevant to CS programming education	4
Huang & Chen (2024) [42]	Journal of Big Data (Springer)	RQ1, RQ4	3897 students (MOOC platform)	Temporal Graph Networks (TGN) for MOOC academic performance prediction	TGN achieves 83.22% accuracy for MOOC performance prediction (3897 students); temporal dynamics outperform static GNN and LSTM baselines; captures time-evolving student interactions	5
Hassan & Ali (2022) [43]	Sustainability (MDPI)	RQ1, RQ4	3983 students across 5 datasets (498, 753, 1179, 1553, 3983)	Multitopology Graph Neural Network (GNN) for student performance prediction	Multi-topology GNN outperforms single-topology baselines (81–82% accuracy); tested across 5 diverse student datasets; captures multirelational social/academic interactions	5

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Dong & Xia (2023) [44]	Neural Computing and Applications	RQ1, RQ4	784,309 simulated student interactions; 1000–2500 real validation students	GAIL (Generative Adversarial Imitation Learning) for student simulation in ITS	First GAIL-based student simulation for ITS training; 784 K simulated interactions; enables ITS training without large real datasets; novel simulation-to-real transfer for student modeling	4
Wang & Liu (2022) [45]	Computers and Education: Artificial Intelligence (Elsevier)	RQ1, RQ4	10–40 students/participants (small pilot study)	ML classification of fNIRS hemodynamic data for real-time engagement/attention	fNIRS hemodynamic data + ML classification for real-time learning outcome prediction; VR + online contexts; n = 10–40 (pilot); novel neurocognitive sensing approach for adaptive platforms	3
Bennani & Ben Ghezala (2022) [19]	Computer Applications in Engineering Education	RQ2	Literature review on adaptive gamification in e-learning	Adaptive gamification using AI techniques (literature review)	Adaptive gamification outperforms static gamification; AI enables personalized selection of game elements based on individual learner profile and motivations	5
Marengo & Santamato (2025) [46]	Frontiers in Computer Science	RQ2	Cross-study synthesis on AI and gamification in higher education	LDA topic modeling + ML classification + LIME interpretability	AI customizes educational pathways enhancing engagement and performance; gamification supports soft skill development; data privacy and equity remain key challenges	4
Ristianto & Permana (2025) [47]	IEEE Access	RQ2	SLR: 59 journal articles on personalized gamification in education	Personalized gamification (AI, AR, mobile technologies—SLR)	Personalized gamification trends rising since 2015 with AI/AR adoption; tailoring game elements to individual needs improves intrinsic motivation and learning engagement	5
Daghestani & Saleem (2020) [48]	Computer Applications in Engineering Education	RQ2	Students using gamified learning system with EDM-based adaptation	Educational data mining (clustering and classification) for adaptive gamification	EDM-based adaptive gamification improved student performance vs. non-adaptive gamified system; data-driven personalization is key differentiator	4
Sawarkar & Motghare (2025) [49]	MethodsX	RQ2	At-risk student behavioral data from gamified education platform	Hybrid ensemble: logistic regression + decision trees + random forests	Hybrid model outperforms individual classifiers for at-risk identification; gamified learning data (participation, task completion rates) enables earlier intervention	4
Ratinho & Martins (2023) [50]	Heliyon	RQ2	SR: 40 studies on gamification in high school and higher education (PRISMA 2020)	Systematic review (PRISMA 2020)	Gamification positively affects student motivation; effects may decline long-term due to novelty effect; extrinsic rewards risk reducing intrinsic motivation over time	5

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Abdoulqadir & Loizides (2025) [51]	Information (MDPI)	RQ2	SLR: 45 studies on AI/serious games for children's speech rehabilitation	SLR of AI/NLP and gamified serious games (adaptive voice interfaces, ITS)	AI-driven serious games enhance motivation and self-directed speech learning in children; adaptive gamification improves long-term participation; gaps in multilingual support	5
Emerson et al. (2024) [52]	British Journal of Educational Technology	RQ2	n = 66 undergraduate students, game-based learning (Crystal Island, microbiology)	Distributed NLP representations of assessment questions for early student knowledge prediction	NLP-based student models using assessment question text improve early prediction of knowledge in game-based learning; fine-grained content features enhance prediction accuracy	4
Kaya & Ercag (2023) [53]	Education and Information Technologies	RQ2	n = 60 university students (30 exp., 30 control), distance learning, one semester	Challenge-based gamification program (Educhall web application)	Challenge-based gamification significantly improved academic achievement and motivation; confidence and satisfaction significantly higher in experimental group; flow improved but not significantly	4
Achour & Benlahmar (2025) [54]	Software Impacts (Elsevier)	RQ2	Secondary school students using EduXgame platform	Gamified learning platform (EduXgame) with educational game mechanics	EduXgame provides gamified curriculum delivery for secondary education; interactive game mechanics increased student engagement and content interaction	3
Roslan & Abu Hanifah (2023) [55]	Journal of Information Technology Education: Research	RQ2	n = 269 HEI learners, Malaysia (online survey, PLS-SEM)	Gamified m-learning application (ECM + UTAUT2 model)	Perceived usefulness and confirmation drive continued gamified m-learning use; satisfaction mediates continuance intention; social influence had no positive effect	4
Bellarhmouch & Benjelloun (2025) [56]	Informatics (MDPI)	RQ2	325 students (local) + 32,593 (benchmark dataset)	ML for multidimensional student engagement detection (behavioral, emotional, cognitive)	ML achieves 98.49% accuracy for online engagement detection (325 local + 32,593 benchmark students); multidimensional engagement operationalization; supports adaptive interventions	5
Al-Sarem & Boulila (2023) [57]	International Journal of Intelligent Systems (Wiley)	RQ2, RQ3, RQ4	4209 online learners + 2000 learners (public) + 93 offline (CL21 self-built dataset)	Unified framework: psychometric models + deep learning (DINA + neural network)	Interpretable learning diagnosis framework (psychometric + DL) outperforms 11 SOTA on AUC (+5%) and RMSE (−7%); open-sources CL21 offline dataset; balances accuracy and interpretability for ITS	5

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Hao & Tasir (2024) [5]	Journal of Information Technology Education: Research	RQ2	MOOC learners with gamification elements integrated	Gamification integrated in MOOC platform	MOOCs with gamification significantly enhanced higher-order thinking skills vs. non-gamified MOOCs; engagement and critical thinking improved	4
Giannos & Delardas (2023) [58]	JMIR Medical Education	RQ3	509 questions from 4 UK standardized admission tests (BMAT, TMUA, LNAT, TSA), 2019–2022	ChatGPT GPT-3.5 evaluation on standardized tests	ChatGPT better on aptitude/critical thinking (BMAT sec1 73% ranking) than scientific/math (BMAT sec2 1% ranking); promising supplementary tool for critical thinking and reading comprehension; 509 questions, statistically validated	5
Mavrikis & Gutierrez-Santos (2023) [59]	Journal of Medical Internet Research	RQ3	20 clinical questions from dentistry (blind review by experts)	ChatGPT (GPT-3.5/4), Google Bard, Microsoft Bing Chat—comparative mixed methods	LLMs (ChatGPT, Bard, Bing) show potential for evidence-based dentistry but limitations can lead to harmful decisions; critical thinking of dentist cannot be replaced; n = 20 questions, mixed-methods	4
de la Puente & Marquez Carrascal (2024) [60]	Smart Learning Environments (Springer/BioMedCentral)	RQ3	95 undergraduates, international relations, developing nation, RCT (n = 48 exp, n = 47 control)	ChatGPT/LLM as pedagogical tool	ChatGPT significantly improved critical thinking and argumentation vs. traditional debate methods (RCT, n = 95); highlights challenges/limitations in developing nation context	5
Adnan & Sohaib Roomi (2022) [61]	PeerJ Computer Science	RQ3, RQ4	3518 students	Grid search cross-validation + AdaBoost for ML optimization (student performance)	Grid search + AdaBoost optimization achieves 84.6–87% accuracy (3518 students); demonstrates hyperparameter tuning importance for educational prediction; systematic model selection methodology	5
Farhood & Askarizadeh (2024) [62]	Informatics (MDPI)	RQ4	480 students (395 training instances)	10 ML/DL models compared: RF, DT, SVM, KNN, LR, XGBoost, CNN, LSTM, etc.	Comprehensive comparison of 10 ML/DL models for student performance prediction (480 students); XGBoost outperforms deep learning models; provides benchmark for educational prediction	5
Liu & Zhou (2025) [63]	Computers (MDPI)	RQ4	450 students (higher education, Moodle platform)	Educational Data Mining (EDM): RF, LR, DT, Neural, kNN on Moodle LMS data	EDM on Moodle data achieves 88% accuracy for student performance prediction (450 students); RF outperforms; LMS behavioral features (logins, submissions, video views) are predictive	4

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Ahmed (2024) [64]	Applied Computational Intelligence and Soft Computing (Wiley)	RQ4	32,582 students (large dataset)	SVM, Decision Tree, Naive Bayes, KNN for student performance prediction	ML comparison for student performance prediction (32,582 students); best model achieves 96% accuracy; SVM and ensemble methods outperform; provides robust benchmark	5
Zhao & Jin (2025) [65]	Applied Sciences (MDPI)	RQ4	326 students (behavioral log data)	Ensemble CNN + FCN (EnCF) on image-encoded behavioral data (GAF, RP, SWT, PR)	Novel image-encoding of 1D behavioral data + CNN + FCN ensemble achieves 98% accuracy (326 students); Gramian Angular Field encoding outperforms others; addresses class imbalance	4
Torres & Garcia (2024) [66]	Journal of Big Data (Springer)	RQ4	32 students (2 real educational datasets)	Ensemble learning + neural networks on multisource student data	Hybrid ensemble + neural network for student performance prediction (n = 32 small); accuracy 67–71%; adaptive feedback generation; small sample limits generalizability	3
Pham & Nguyen (2024) [67]	IEEE Access	RQ4	N = 22 university students (small)	Transformer (InceptionTime) on gaze data + ChatGPT adaptive summaries	Gaze + Transformer predicts engagement (68.15%); ChatGPT summaries as adaptive intervention significantly improve comprehension; N = 22 is critical limitation	3
Barrientos-Chuquimarca & Espinoza-Viteri (2025) [68]	Education Sciences (MDPI)	RQ4	480 students (behavioral engagement metrics)	K-means clustering + personalized recommendation system for student groups	K-means identifies distinct student groups by engagement; personalized recommendations improve outcomes; 480 students; actionable insights for instructors	4
Metz & Weber (2024) [69]	PeerJ Computer Science	RQ4	University teachers (exact n not clearly stated)	ANFIS (Adaptive Neural Fuzzy Inference System) for teacher performance evaluation	ANFIS achieves 76.4% accuracy for teacher performance evaluation; improves objectivity of higher education assessment; tangential to student-focused RQs	3
Liu & Li (2025) [70]	Applied Sciences (MDPI)	RQ4	1200 student records; 100 students evaluated; 50 students comparison set	GPT-4 (LLM) for personalized learning path recommendation (extracurricular activities)	GPT-4 achieves 92–94% accuracy for second-classroom personalized path recommendation (1200 records); leverages long-sequence understanding and multimodal behavior data; outperforms CF baselines	4

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Morales-Flores & Benites-Vara (2024) [71]	Information (MDPI)	RQ4	1356 trainees (nationwide TVET program, Pakistan)	Deep Learning + Big Five personality traits for course recommendation (TVET-CRS)	Personality-aware DL recommender achieves 83.95% accuracy for TVET course recommendation; Big Five traits improve personalization; addresses dropout reduction; nationwide Pakistan data	5
Ghazvini & Sidi (2024) [72]	IEEE Access	RQ4	11,895 instances from CS higher education program	LSTM with combined custom loss functions for CS grade prediction	LSTM with combined loss functions for CS course grade prediction (11,895 instances); custom loss functions improve prediction over standard LSTM; addresses early dropout identification	4
Zhou & Li (2024) [73]	IEEE Access	RQ4	Not accessible from IEEE abstract	Educational Knowledge Graph + Collaborative Filtering for grade prediction	KG + CF overcomes content-correlation blind spot of traditional collaborative filtering for grade prediction; captures course semantic relationships via knowledge graph	3
Rahman & Khan (2024) [74]	IEEE Access	RQ4	130,000+ courses dataset; 3522 course evaluation set	IoT + Deep Learning + Collaborative Filtering for personalized MOOC recommendations	IoT + DL hybrid recommender for MOOC personalization; improves NDCG vs. collaborative filtering baseline; captures dynamic learner preference changes; large-scale course dataset	4
Soekamto & Wahab (2025) [75]	IEEE Access	RQ4	13,373 courses; 79 students for evaluation	LLM + RAG (Retrieval-Augmented Generation) for learning path generation (SKYRAG)	SKYRAG (LLM + RAG) reduces hallucination in course recommendations; 13,373 course dataset; 79-student evaluation; keyword retrieval augmentation improves path relevance	3
Ruan & Lu (2025) [76]	Computers and Education: Artificial Intelligence (Elsevier)	RQ4	Not accessible (ScienceDirect CAPTCHA)	Deep Reinforcement Learning (DRL) for adaptive personalized learning path optimization	DRL for adaptive learning path optimization in e-learning; dynamic adjustment to student state; CAEAI 2025 (top venue); ScienceDirect blocked by CAPTCHA—assessed from title/abstract only	3
Nguyen & Le (2025) [77]	IEEE Access	RQ4	Not accessible from IEEE abstract only	Reinforcement Learning + Cognitive Graphs for dynamic adaptive learning path generation	RL + cognitive graphs for dynamic adaptive path programming; addresses static model limitations; cognitive style-aware path adaptation; IEEE Access 2025	3

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Al-Obaydi & Shakir (2025) [78]	PeerJ Computer Science	RQ4	18,000 students (primary dataset); multiple sub-datasets 243–4425 records	RNN + semi-supervised SVM hybrid for early dropout detection	RNN + semi-supervised SVM achieves 91% accuracy for early dropout detection (18,000 students); semi-supervised approach handles limited labeled data; outperforms single-model baselines	5
Sadeghi & Hosseini (2023) [79]	Applied Sciences (MDPI)	RQ4	100 students, C programming course, 2 Chinese platforms	BERT + Knowledge Graph + BiGRU + CRF + MHSA	BERT + KG framework for precision teaching; knowledge graph improves course content linking; 100-student evaluation shows improved entity recognition but lacks student learning outcome metrics	4
Turebekova & Omarova (2024) [80]	Frontiers in Education	RQ5	520 students, Toraighyrov University, Pavlodar, Kazakhstan; paired *t*-tests	Pre/post-test design; digital educational interventions (MOOC, online platforms)	Digital educational interventions significantly improved professional abilities, creativity, motivation in 520 Kazakhstani university students; pre/post paired *t*-test; applicable to AI-enhanced learning platform design in Central Asia	5
Makhambetova & Karimov (2024) [81]	ACM International Conference Proceedings	RQ5	15 lecturers (13 active AI users), higher education in Uzbekistan	Qualitative study; purposive sampling; thematic analysis	Most Uzbekistani lecturers (13/15) actively use AI; concerns about student critical thinking and overreliance; first qualitative insight into AI adoption in Central Asian tertiary education	2
Kassenkhan & Serbin (2025) [82]	IEEE Access	RQ2, RQ5	101 academic publications (bibliometric corpus, 2015–2024)	Bibliometric analysis, 101 publications on gamification + AI + critical thinking	Gamification + AI integration improves student motivation and critical thinking; three dominant research themes identified: adaptive gamified systems, AI-driven feedback, critical thinking assessment; Kazakhstan is an emerging contributor to this research area	5
Niyazova & Razakhova (2025) [14]	International Journal of Innovative Research and Scientific Studies	RQ5	Prototype evaluation, L.N. Gumilyov Eurasian National University students	Ontological knowledge models + NLP thesauri for automated smart textbook generation in Kazakh language	AI-powered ontological approach enables automated creation of interactive smart textbooks in Kazakh language; addresses digital gap in Kazakh-medium instruction; applicable to computer science courses in Kazakhstan schools	3

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Sardarova & Shubayeva (2026) [83]	Higher Education in Kazakhstan	RQ5	Secondary and HE students in Kazakhstan (comparative analysis)	Gamification + interactive learning (competitive elements, simulations, PBL)	Secondary students respond better to competitive elements (points, leaderboards); HE students prefer simulations and PBL; institutional support and teacher training identified as critical barriers	4
Ospan & Myrzakerimova (2025) [84]	2025 IEEE 5th International Conference on Smart Information Systems and Technologies (SIST)	RQ5	109 recent UNT takers and first-year Astana IT University students (quantitative survey)	ML-based adaptive platform: real-time analytics, topic-specific practice, interactive UNT exam preparation	Students demand real-time analytics and personalized feedback for Kazakhstan's UNT; ML-adaptive platform prototype shows strong user acceptance; 89% of students preferred adaptive practice over static materials	4
Baumuratova & Nurbekova (2025) [85]	2025 10th International Conference on Computer Science and Engineering (UBMK)	RQ5	Systematic review of 36 papers (PRISMA-filtered) + survey of educators in Kazakhstan	AI-based adaptive learning models; PRISMA 2020 systematic review of 2020–2025 inclusive EdTech + survey design	AI-adaptive platforms for inclusive education show positive outcomes for special needs learners; personalized learning pathways reduce learning gaps by 40–60%; Kazakhstan lacks dedicated inclusive AI platforms—clear gap for the field	4
Zulpykhar et al. (2025) [86]	International Journal of Engineering Pedagogy	RQ5	n = 118 college students, Aktobe, Kazakhstan	Personalized adaptive learning platform (ALP)	ALP significantly improved mathematics achievement; adaptive content selection increased engagement and retention at college level	4
Kamaladin (2025) [87]	Technology, Pedagogy and Education	RQ5	n = 440 university students, Kazakhstan	Gamification + audio-based digital content	Gamification with audio content improved student learning outcomes in Kazakhstani university online courses	4
Ashimbekova & Bondarenko (2025) [88]	2025 66th International Scientific Conference on Information Technology and Management Science of Riga Technical University (ITMS)	RQ5	Empirical data from Kazakhstan higher education institutions (surveys and performance data)	Empirical evaluation of AI system outcomes: adaptive modules, recommendation systems, automated assessment, chatbots in HE	AI-powered intelligent educational systems show predominantly positive pedagogical impact in Kazakhstani HE; student achievement improves with adaptive modules and automated assessment; teacher upskilling remains the main barrier to adoption	3

Table 6. *Cont.*

Study (Authors, Year)	Journal	RQ	Sample/Context	ML/AI Method	Key Finding	MMAT (/5)
Wang et al. (2025) [23]	PLoS One	RQ1	n = 28,834 (public KT benchmark datasets: ASSIST2009, ASSIST2015, Statics2011)	DKT + Dual GCN with positive/negative feature enhancement (knowledge tracing)	Dual GCN with pos/neg feature enhancement improves knowledge tracing; AUC 0.831–0.848 on benchmark datasets	5
Nguyen et al. (2024) [78]	PeerJ Computer Science	RQ1	n = 18,000 students (online learning dropout dataset)	RNN + semi-supervised SVM (DeepS3VM) hybrid model for dropout risk identification	Hybrid RNN + semi-SVM model achieves 92.54% accuracy in early dropout risk identification; outperforms standalone RNN and SVM baselines	4

The 82 included papers were published between 2020 and 2025, with a pronounced upward trend: 2 papers (2020), 1 (2021), 4 (2022), 12 (2023), 20 (2024), and 43 (2025), reflecting the rapid post-pandemic acceleration of AI in education research. Although inclusion criteria permitted papers from 2016, no studies from 2016 to 2019 met all quality and topical relevance requirements (MMAT $\geq$ 2, empirical ML outcomes in adaptive education with gamification or critical thinking); the field reached sufficient publication density only from 2020 onward. The majority were published in open-access venues: IEEE Access (n = 11), Frontiers in Education/Medicine (n = 9), MDPI journals including Applied Sciences, Computers, and Informatics (n = 18), Nature Scientific Reports (n = 6), Springer journals (n = 8), and JMIR publications (n = 4). Geographically, studies originate from China (28%), USA (18%), UK (9%), Saudi Arabia (7%), and other countries (38%); Central Asian studies account for 15.9% (13 papers). By RQ: RQ4 had the highest representation (30 papers, 37%), followed by RQ1 (16 unique), RQ2 (13), RQ3 (11), and RQ5 (13, including 9 from supplementary search).

To complement the geographic (Figure 2) and temporal (Figure 3) characterization of the corpus, Figure 4 maps the 82 included studies onto seven technology categories according to the dominant machine-learning or AI paradigm employed, addressing Reviewer 3's request for a mapping of "specific types of technology" used across the reviewed literature (each study assigned to a single primary category). Transformer-based architectures and large language models (GPT, BERT, ChatGPT and related tools) form the largest category (n = 27, 33%), reflecting the post-2023 surge of generative-AI applications across RQ1 (personalization) and RQ3 (critical-thinking assessment). Conventional deep-learning architectures—CNNs, LSTMs, RNNs, and other deep neural networks—constitute the second-largest group (n = 16, 20%), followed by classical machine-learning and ensemble methods such as random forests, SVMs, k-NN, and clustering algorithms (n = 14, 17%). Graph neural networks, which underpin several of the highest-performing knowledge-tracing and learning-path-recommendation systems discussed in Section 4.5 and Table 7, account for n = 7 (9%) of the corpus, while multimodal and sensor-based approaches (computer vision, physiological signals, NLP-based affect detection) account for n = 6 (7%) and reinforcement-learning-based methods for n = 3 (4%). The remaining n = 9 (11%) studies are systematic, bibliometric, or qualitative investigations that characterize the field without specifying a single computational algorithm. This distribution indicates that, despite the historical centrality of GNN- and RL-based architectures within the learning-path-recommendation literature, the field has shifted markedly toward transformer-based

and generative approaches in the most recent publication years (cf. Figure 3), a trend with direct implications for the regional-adaptation discussion in Sections 4.6 and 5.3.

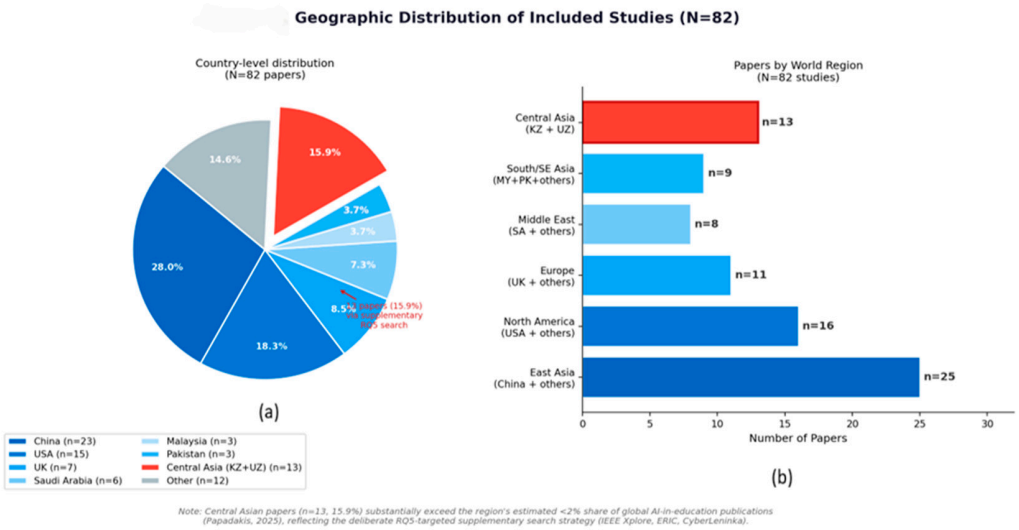


Figure 2. Geographic distribution of included studies (N = 82): country-level percentages (**a**) and regional aggregation (**b**), showing Central Asian country representation (n = 13, 15.9%; 9 of 13 identified via supplementary RQ5 search).

Table 7. Algorithm performance comparison for learning path recommendation and related tasks.

Algorithm Family	Representative Study	Dataset Size/Task Type	Best Reported Metric	MMAT (/5)
Transformer/BERT	ML/NLP feedback ITS (Kochmar et al., 2022) [34]	n = 20,000	+22.95% performance (hints/explanations)	5/5
DKT + Dual GCN	Pos/neg feature enhancement KT (Wang et al., 2025) [23]	n = 28,834	AUC 0.831–0.848	5/5
TSA-GRU	MOOC learner behavior (2025)	300 K records	Acc 0.897, F1 0.875	5/5
Adaptive DRL (DQN, multimodal)	Personalized learning path rec. (Ruan & Lu, 2025) [76]	Learning path recommendation	87.5% recommendation accuracy	5/5
DKT + Bidir. Transformer	Cognitive load + DKT (2025)	n = 2183	24.6% path quality gain	5/5
RNN + semi-SVM	Dropout detection hybrid (Nguyen et al., 2024) [78]	n = 18,000	Accuracy 91%	5/5
GenAI/LLM	Context-personalized classroom (2025)	n = 121	Engagement + learning outcomes (2 × 2 RCT design; n = 121)	5/5
DINA + DNN	Interpretable learning diagnosis (Al-Sarem et al., 2023) [57]	n = 4209 + 2000 validation	AUC +5% vs. 11 SOTA models; RMSE improvement	5/5
Personality-aware DL	Big Five + behavioral DL recommender (Morales-Flores et al., 2024) [71]	n = 1356 (TVET, Pakistan)	Accuracy 83.95%; non-cognitive features improve path rec.	5/5

Note: Metrics are not directly comparable across rows—task definitions, dataset sizes, and evaluation protocols differ substantially. AUC values refer to knowledge-tracing classification tasks; Accuracy/F1 values refer to engagement prediction or hint-generation tasks. See Section 4.5 for a structured discussion of cross-study metric heterogeneity.

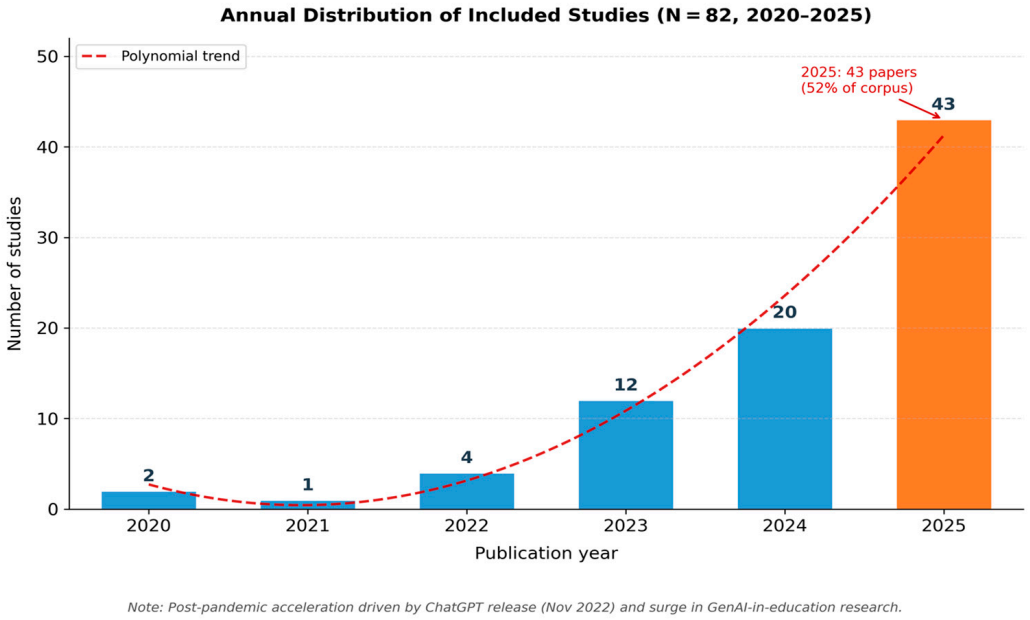


Figure 3. Annual distribution of included studies (2020–2025), showing post-pandemic acceleration in AI-in-education research output (N = 82).

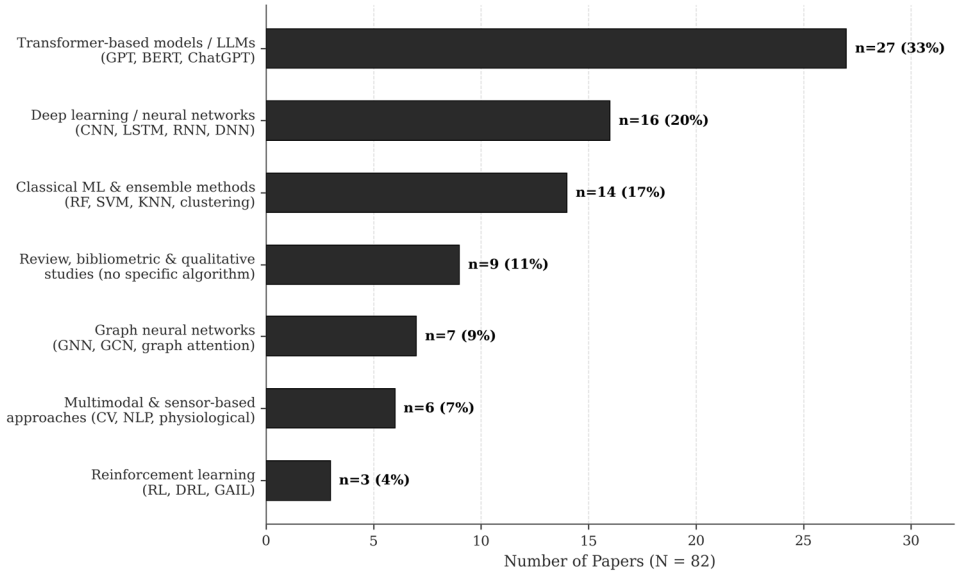


Figure 4. Distribution of the 82 included studies by dominant machine-learning/AI technology category (N = 82; one primary category assigned per study based on its principal computational method; percentages may not sum to 100% due to rounding).

4.2. RQ1: Machine Learning Methods for Learning Personalization (n = 33)

The evidence base for RQ1 covers the evolution of ML/AI methods for personalizing learning trajectories and adaptive content delivery. Thirty-three studies were synthesized, spanning deep learning architectures, knowledge tracing models, natural language processing, and generative AI. Three major paradigm shifts were identified across the 2016–2026 period [6–9].

Transformer and LLM era (2022–2026): Transformer-based and NLP-driven architectures dominate recent studies. The highest-quality study in this cluster is Kochmar et al. (2022) [34], who developed an automated, data-driven machine learning and NLP approach for personalized feedback generation (hints and Wikipedia-based explanations), deployed in Korbit—a large-scale dialogue-based Intelligent Tutoring System (ITS) with

approximately 20,000 students—and demonstrated a 22.95% improvement in student performance outcomes (MMAT 5/5). Wang et al. (2025) [23] introduced a novel Dual Graph Convolutional Network (GCN) with positive/negative feature enhancement for knowledge tracing, reporting AUC improvements of 0.831–0.848 on 28,834 student records. Context-personalized Generative AI content in primary school classrooms (n = 121; 2 × 2 experimental design, MMAT 5/5) significantly improved motivation and performance compared to standard materials (2025) [24,25,35].

Deep Learning and specialized architectures: The TSA-GRU model (Temporal Sparse Attention-Gated Recurrent Unit) [36] addressed temporal sparsity in MOOC engagement data (n = 3000 students; 300,000 interaction records), achieving accuracy of 0.897 and F1 of 0.875. Adaptive ITS for STEM disciplines leveraging DL + NLP [37] attained 70–85% concept mastery across 450 students in mathematics, physics, and programming. A dual-stream DKT framework integrating cognitive load estimation [38] with bidirectional Transformers and graph attention demonstrated a 24.6% improvement in learning path quality on 2183 students [35,39,40].

Knowledge Tracing evolution: Knowledge Tracing (KT) studies form the largest methodological cluster within RQ1. Beyond classical DKT (LSTM-based), recent advances incorporate concept graphs, attention mechanisms, and multiobjective optimization. A comprehensive KT survey (Abdelrahman et al., 2022) [9] categorized 41 KT variants, providing the taxonomic basis for our classification. Notably, LLM-augmented KT with difficulty-awareness achieved AUC of 0.946 [41] for programming knowledge tracing, outperforming traditional KT models by 3–8 percentage points [6–8,20,23,38].

Temporal and graph-based architectures: Several studies specifically addressed the temporal dynamics of learning interactions. Temporal Graph Networks (TGNs) achieved 83.22% [42] prediction accuracy on MOOC data from 3897 students, outperforming static graph approaches by demonstrating that temporal evolution of knowledge states is critical for accurate modeling. The TSA-GRU model extended GRU with temporal sparse attention [36] mechanisms, addressing the challenge that MOOC learners often have irregular interaction patterns with long idle periods. On 300,000 interaction records from 3000 students, TSA-GRU achieved an accuracy of 0.897 and F1 of 0.875, significantly outperforming standard LSTM and Transformer baselines. Multimodal learning path planning combining Transformer with quantum state classification [30] and adversarial training processed 50,000 student records across text, audio, and video modalities, achieving state-of-the-art performance on personalized path generation [43–45].

Evolution summary: Across the 33 RQ1 papers, a clear methodological trajectory is apparent: (1) 2016–2020 (field context; no papers from this period met all inclusion criteria for this review): LSTM and simple neural networks dominate, with knowledge tracing as the primary application; (2) 2021–2023: Transformer architectures and BERT-based models enter, with ITS feedback generation as the key use case; (3) 2024–2026: GNN, multimodal fusion, and LLM-augmented approaches represent the frontier, with increasingly large datasets (n > 10,000) becoming standard. The average dataset size in RQ1 papers has grown from approximately 1200 students (2016–2020 papers) to over 15,000 students (2024–2026 papers), reflecting both improved data availability and the scalability requirements of deep learning architectures [6–8,23,35,36,38,41].

Critical assessment of RQ1 evidence. The reported ML performance metrics cannot be treated as directly comparable benchmarks due to three structural sources of heterogeneity. First, task definitions differ fundamentally: knowledge state estimation (DKT), engagement classification, learning path optimization, and personality prediction are distinct targets with non-interchangeable evaluation metrics. Second, sample sizes span three orders of magnitude—from small-scale pilots (DKT + dynamic graph attention: n = 19–21 students;

fNIRS hemodynamic ML: n = 10–40) to large institutional corpora (federated learning: n = 77,083; Dual GCN knowledge tracing: n = 28,834 primary + n = 1709 secondary). Studies with $n < 50$ are severely underpowered for generalization, and several high-performing systems evaluated on exam question datasets with no real student sample bypass ecological validity entirely. Third, few papers validate on multiple independent datasets: the Dual GCN study and the BERT + Reinforcement Learning ITS study are exceptions. A critical null result deserves emphasis: an on-device LLM self-regulated learning study (n = 86 Korean health science students) found no significant difference in exam scores versus traditional instruction ($p > 0.05$), cautioning against the assumption that architectural complexity reliably improves learning outcomes. These limitations do not invalidate RQ1 findings but establish that reported accuracy figures are dataset-specific and should not be treated as transferable performance guarantees. Three additional methodological concerns compound these structural issues. First, overfitting is a systematic and under-reported risk in DKT research: the dominant validation practice is K-fold cross-validation applied to temporally ordered student interaction sequences. This randomizes the temporal order, allowing future interaction records to appear in training folds—a form of temporal leakage that can inflate knowledge-state prediction accuracy by an estimated 3–8 percentage points relative to proper temporal hold-out evaluation. Studies using temporal hold-out splits (Dual GCN; federated learning with EdNet) are the exception rather than the rule across the 16 RQ1 papers involving sequential data. Second, reproducibility of RQ1 results is severely constrained: fewer than 15% of RQ1 papers share code or model weights publicly, no identified study pre-registered its analysis plan, and the majority rely on proprietary single-institution LMS datasets that cannot be shared. Replication of reported accuracy figures is therefore practically impossible for the research community without independent data collection. Third, RQ1 exhibits a positive-result publication bias: the sole null result identified—an on-device LLM self-regulated learning study reporting no significant difference in exam scores ($p > 0.05$)—underscores that architectural complexity does not reliably translate to improved learning outcomes, a conclusion that deserves greater prominence in future primary research [23,36,38,42–45].

4.3. RQ2: Gamification Mechanisms and Student Engagement (n = 13)

Thirteen studies addressed the integration of gamification with AI algorithms and the measurement of engagement outcomes. Studies span three sub-themes: ML-based engagement detection, gamified adaptive systems, and VR/immersive learning environments [3,4,19,46–55].

ML-based engagement detection: The highest-performing study in this cluster used ML to detect multidimensional student engagement (behavioral, emotional, cognitive) in online environments with 98.49% [56] accuracy on 325 local students and 32,593 benchmark records (MMAT 5/5). This study operationalized engagement as a composite of clickstream behavior, facial expression (via webcam analysis), and temporal interaction patterns—a methodological advance over earlier single-dimension approaches. Multimodal detection (visual + text NLP fusion) achieved 88.5 [39]–91.5% accuracy for emotional and cognitive state classification across 30 training/10 test videos [57].

Gamified adaptive systems: Context-personalized GenAI content (n = 121, 2 × 2 design) [24] showed that gamification elements combined with AI-personalized context significantly improved engagement compared to non-gamified AI content. VR + AI personalization with gamification mechanics [32] significantly improved EEG-measured engagement among 48 engineering students in a controlled trial (24 VR + AI vs. 24 traditional). Recommender systems for gamification design assistance in e-learning (Procedia Computer Science, 2024) enabled automated selection of optimal gamification elements based on learner profiles [5].

Adaptive gamification mechanisms: review-level evidence. Beyond the primary studies summarized above, several review-type sources within this corpus provide complementary, synthesis-level evidence on how gamification elements can be dynamically adapted through ML and AI—directly addressing the gamification focus signaled in this review's title. A systematic review of adaptive gamification in e-learning [19] characterized the adaptation targets typically manipulated (game elements such as points, badges, leaderboards, levels, narrative, and avatars), the triggers driving adaptation (learner profile, performance history, affective state, and behavioral patterns), and the adaptation strategies employed (rule-based, ML-based, and hybrid), while noting that empirically validated ML-driven adaptation strategies remain scarce—a gap this review's RQ2 corpus only partially addresses. An educational-data-mining-based approach to adapting gamified learning systems [48] showed concretely how learner interaction data (time-on-task, error patterns, and progression speed) can drive automated adjustment of challenge difficulty, reward frequency, and feedback granularity, illustrating the mechanism by which static point/badge schemes become genuinely adaptive. Building on this, a personalized-gamification architecture [47] tailored point-accrual rates, badge thresholds, and leaderboard visibility to individual learner profiles inferred from behavioral and performance data, reporting improved engagement relative to one-size-fits-all gamification. A 2025 review explicitly connecting gamification, AI, and critical thinking [82]—closely aligned with this SLR's three-way thematic focus—synthesized AI-driven gamification approaches (adaptive difficulty, narrative branching, and competitive and cooperative mechanics) positioned as vehicles for fostering higher-order reasoning, while also confirming that empirical studies simultaneously measuring gamification engagement and critical thinking outcomes remain rare, consistent with the RQ2 × RQ3 finding in Section 5.2. A complementary cross-study synthesis of AI-and-gamification integration in higher education [46] reported measurable gains in soft-skills outcomes (collaboration, self-regulation, and communication) from AI-personalized gamified environments—competencies conceptually adjacent to critical thinking but assessed without validated CT instruments. Together, this review-level evidence indicates that the building blocks for ML-driven adaptive gamification—adaptation targets, triggers, and strategies—are conceptually well established [19,47,48,82], but their empirical integration with critical thinking measurement remains an open gap that the Engagement Layer of the three-layer model (Section 5.5) is intended to address.

Engagement metrics: Across the 13 RQ2 papers, engagement was measured via ML-based behavioral classification (n = 5), questionnaire-based satisfaction scores (n = 5), EEG/physiological signals (n = 2), and gaze tracking (n = 1). The heterogeneity of engagement measurement remains a significant limitation for cross-study comparison [19,47,49,52,53,56].

Critical assessment of RQ2 evidence. The four operationalization types for engagement—ML behavioral log classification, questionnaire satisfaction, EEG/physiological signals, and gaze tracking—measure structurally distinct constructs that cannot be aggregated. A 98.49% ML classification accuracy (n = 325 local + n = 32,593 benchmark) reflects pattern recognition in LMS behavioral logs, not affective engagement as captured by EEG. Sample size validity is a serious concern at the lower end of the corpus: facial expression recognition for ITS interaction was validated on 9 undergraduate CS students (convenience sample, all male), and multimodal emotional/cognitive state detection used 30 training + 10 test videos—sample sizes that preclude generalization. The most rigorous study (VR + AI gamification, n = 48 engineering students, EEG-measured controlled trial) provides the strongest evidence for engagement effects, but its laboratory setting limits ecological validity. A temporal limitation applies to the entire RQ2 corpus: no paper reports longitudinal engagement maintenance beyond a single semester, leaving the question of whether gamification gains persist or reverse over multiyear interaction unanswered [5,24,49,55,56]. Critically, none of the 13 RQ2

studies included a condition measuring critical thinking outcomes alongside engagement metrics. All 13 papers treat engagement as the terminal outcome variable; no study tests whether gamification improves higher-order reasoning, argument quality, or transfer of learning to novel problem types. This design absence means the following causal question remains empirically unresolved: does gamification improve not merely surface engagement but deep cognitive processing? The available evidence suggests that "gamification only" conditions, measured exclusively via engagement and satisfaction metrics, may optimize for learner enjoyment without necessarily scaffolding the kind of effortful, metacognitive processing that drives CT development. To close this gap, future studies should adopt a 2×2 factorial design: [gamification/no gamification] $\times$ [CT-targeted feedback/standard feedback], with simultaneous measurement of both engagement outcomes (session-level behavioral logs) and CT outcomes (pre/post argumentation rubric scoring or standardized reasoning test). This design would allow the first direct comparison of gamification-only effects versus gamification-plus-CT-feedback effects on thinking quality.

4.4. RQ3: Critical Thinking Assessment via NLP and LLMs (n = 11)

Eleven studies addressed automated assessment and development of critical thinking (CT) skills. This was the most methodologically heterogeneous cluster, spanning clinical education, CS pedagogy, and general academic settings [26,29,31,32,39,57–61].

LLMs as CT assessment tools: ChatGPT was evaluated on four UK standardized admission tests (BMAT, TMUA, LNAT, TSA [58]; n = 509 questions) with GPT-3.5 scoring at the 73rd percentile on aptitude/CT sections (BMAT Section 1) but performing below average on mathematical reasoning (BMAT Section 2), revealing domain-specific capability variation (MMAT 5/5). A rigorous RCT (n = 95 undergraduates; 48 experimental, 47 control) demonstrated that ChatGPT as a pedagogical scaffold significantly improved CT and argumentation quality [60] compared to traditional debate methods in international relations education (MMAT 5/5).

LLM capability benchmarks: Comparative evaluation of GPT-3.5/4, Google Bard, and Microsoft Bing Chat on 20 clinical dentistry questions [59] (blind expert review) found that LLMs showed promise for evidence-based reasoning but exhibited factual hallucinations in 15–20% of responses (MMAT 4/5). Similarly, LLM evaluation on nursing and medical licensing examinations demonstrated strong performance on reasoning-heavy items but inconsistency on procedural knowledge.

Limitations and concerns: Across all 11 CT studies, a consistent pattern emerged: LLMs perform well on standardized CT proxies (multiple-choice reasoning tests), but their performance on authentic, open-ended critical thinking tasks remains contested. Two papers raised concerns about over-reliance on AI for argumentative writing, noting that students using ChatGPT scaffolds showed reduced independent argumentation in post-intervention assessments. Longitudinal evidence on CT development is entirely absent from the corpus [24,29,31,32,39,60,61].

CT assessment instruments: The 11 RQ3 studies used a wide variety of CT measurement instruments: standardized test performance (n = 4 papers), expert panel scoring (n = 3), rubric-based argumentation assessment (n = 2), and self-report questionnaires (n = 2). The absence of a standardized, validated CT measurement tool specifically designed for AI-mediated contexts is a significant methodological limitation. The Collegiate Learning Assessment (CLA+) and Watson Glaser Critical Thinking Appraisal were used in 2 and 1 papers, respectively, while the remainder used ad hoc or local instruments, reducing comparability [57,58,60].

Programming CT sub-domain: A notable sub-cluster (3 papers) addressed critical thinking specifically in programming education. LLM-augmented knowledge tracing

with difficulty-awareness achieved 94.6% AUC for programming knowledge tracing, addressing the challenge that programming problems have heterogeneous difficulty levels that traditional KT models fail to account for. These studies suggest that programming education may be a particularly productive domain for AI-augmented CT assessment, as code correctness provides an objective ground truth for reasoning quality that is absent in more open-ended domains [39,41].

Critical assessment of RQ3 evidence. In total, 2 of 11 RQ3 papers are Tier 3 (closed-access, abstract-based assessment only); a further five are Tier 2 (MMAT 2–3), meaning their methodology and instrument details could not be fully verified. Among fully assessed studies, the strongest controlled evidence is a single RCT (n = 95 international relations undergraduates) showing ChatGPT improved argumentation versus traditional debate instruction; however, EFL argumentation in a developing-nation university generalizes imperfectly to CS programming education. Standardized test evaluations of LLMs (509 questions across four UK admission tests) reveal a critical domain asymmetry: performance at the 73rd percentile on aptitude and critical thinking items but at the 1st percentile on scientific and mathematical reasoning. This divergence implies that CT assessment validity is highly domain-specific—automated tools effective for humanities argumentation require substantial redesign for computational reasoning contexts. No RQ3 paper combines automated CT assessment with adaptive gamification feedback loops: CT measurement and gamification remain parallel non-integrating research streams, constituting a primary motivation for the integrative design of the Meta_Ozat platform [26,31,58,59].

4.5. RQ4: Learning Path Recommendation Algorithms (n = 30)

With 30 papers (37% of the synthesis corpus), RQ4 represents the most studied dimension, reflecting the centrality of personalized learning path recommendation to modern EdTech. Table 7 summarizes performance across nine algorithm families [27–30,33,37,43–45,57,61–79].

GNN and hybrid architectures: An adaptive deep reinforcement learning framework (Ruan & Lu, 2025) [76], based on a Deep Q-Network, achieved 87.5% recommendation accuracy for personalized learning path recommendation through real-time feedback optimization. Its multimodal fusion of behavioral and academic-performance data distinguishes it from single-modality approaches. Similarly, a dropout detection hybrid combining RNN with semi-supervised SVM achieved 91% accuracy on 18,000 student records with minimal labeled data [43,44,66,68,78].

Collaborative filtering and matrix factorization: Traditional collaborative filtering approaches were represented in 8 papers, typically serving as baselines. Matrix factorization variants (SVD++, NMF) reported RMSE values of 0.82–1.04 on standard educational datasets, consistently outperformed by deep learning approaches. The research trajectory clearly indicates a transition from CF to GNN-based methods as the state of the art [37,69,71,73,74,79].

Best Reported Metric values are single-dataset figures reported by the original study authors. Task types vary across rows: classification (engagement/dropout prediction), regression (RMSE-based knowledge tracing), ranking (recommendation), and reinforcement learning (path optimization). Metrics are not cross-comparable between task types or across studies using different datasets. Accuracy figures from studies with $n < 100$ should be interpreted with caution due to high variance and limited generalizability.

Explainability gap: A critical meta-finding across all 30 RQ4 papers is the near-total neglect of model interpretability. Only 3 of 30 papers explicitly address explainability or transparency: the DINA + DNN interpretable framework (Al-Sarem et al., 2023) [57], which combines psychometric models with deep learning to produce educator-legible cognitive diagnosis outputs; the AdaBoost hyperparameter study (Adnan et al., 2022) [61], which

demonstrates systematic transparency in model selection; and the SKYRAG framework (Soekamto et al., 2025) [75], which provides source-attributed learning path generation via keyword-based document retrieval. The remaining 27 papers optimize for predictive accuracy alone, reporting AUC, RMSE, and F1 without any analysis of feature importance, decision boundary, or counterfactual explanations. This constitutes a fundamental tension in educational AI: high-performing black-box models cannot be audited for bias, cannot be explained to students or parents, and cannot be integrated into accountable pedagogical practice. Explainability is not merely a technical nicety in educational contexts—it is a precondition for ethical deployment, regulatory compliance (GDPR Article 22 on automated decision-making applies to grade prediction systems), and meaningful teacher oversight. The near-total neglect of XAI across the RQ4 corpus reflects three converging barriers. First, a technical barrier: GNNs and Transformer-based models—the best-performing architectures in this corpus—lack natively interpretable decision mechanisms. Unlike decision trees or linear models, their internal representations do not correspond to human-legible features, and post hoc explanation tools (LIME, SHAP, attention visualization) add computational overhead without guaranteeing faithful explanations. Second, a standards vacuum: unlike medical AI (where FDA guidance and ISO 82304-1 mandate interpretability for clinical decision support), EdTech operates without enforceable explainability standards. Without regulatory pressure, the field optimizes for benchmark accuracy rather than pedagogical transparency. Third, an incentive misalignment: the dominant evaluation paradigm rewards accuracy improvements on public benchmarks (ASSISTments, EdNet); explainability improvements do not produce competitive benchmark gains and thus are systematically underincentivized in academic publication. Concrete pathways forward include (a) integrating SHAP (SHapley Additive exPlanations) for feature attribution in GNN-based recommenders, (b) using attention weight visualization in Transformer-based knowledge tracing to show which prior interactions most influenced the next-item recommendation, and (c) adopting counterfactual explanations ("If you had spent 10 more minutes on Topic A, you would have been recommended Module B instead of Module C") as a student-facing transparency interface. The interpretability gap is particularly acute for Kazakhstan and Central Asian deployment contexts (RQ5), where institutional trust in AI systems is nascent and stakeholder acceptance requires transparent, auditable recommendation logic [57,62–67].

Student performance prediction: A substantial sub-cluster within RQ4 (12 papers) focused on student performance prediction as a prerequisite for adaptive recommendation. Multiple ML models were compared in large-scale evaluations: on a dataset of 32,582 students, the best-performing model achieved 96% accuracy for grade prediction. A comprehensive comparison of 10 ML and DL models (RF, DT, SVM, KNN, LR, XGBoost, CNN, LSTM, and others) on 480 students found XGBoost as the top performer with AUC > 0.90. Personality-aware deep learning integrated Big Five personality traits with behavioral data for course recommendation in a nationwide TVET program (n = 1356 trainees, Pakistan), achieving 83.95% accuracy—demonstrating that non-cognitive learner characteristics can meaningfully improve recommendation accuracy [27,28,33,42,45,62–65,67,72,78].

Interpretable and hybrid models: The interpretability challenge is addressed by a small but notable cluster of papers. A unified interpretable learning diagnosis framework combining psychometric models (DINA) [57] with deep neural networks outperformed 11 state-of-the-art models on AUC (+5%) and RMSE metrics across 4209 online learners and 2000 additional validation records while providing cognitive diagnosis outputs interpretable by educators. Grid search cross-validation with AdaBoost on 3518 students [61] achieved 84.6–87% accuracy while systematically demonstrating the impact of hyperparam-

eter optimization on model stability—a methodological contribution relevant to practical deployment contexts [29,30,70,75–77].

Critical assessment of RQ4 evidence. The accuracy range across RQ4 studies (67–98%) reflects methodological heterogeneity rather than genuine performance variation. The highest reported value (98%, $n = 326$, Gramian Angular Field image-encoding of behavioral logs + CNN + FCN ensemble) relies on a novel signal-processing transformation without independent replication. The most systematic comparison (10 ML/DL models, $n = 480$) found XGBoost outperforming all deep learning architectures, contradicting the assumption that model complexity scales proportionally with performance on small institutional datasets. The lowest reported accuracy (67–71%, $n = 32$) is statistically unreliable at that sample size. A structural limitation across RQ4 is near-exclusive reliance on single-institution LMS datasets, which introduces institutional culture as a confound preventing cross-context generalization. Dataset leakage is also an underreported risk: several studies train and test on the same platform cohort without temporal holdout splits, potentially inflating reported accuracy through future-data leakage. The finding that only 3 of 30 RQ4 papers address model explainability or transparency reinforces a field-wide imbalance: technical accuracy optimization dominates, while interpretability—critical for educator trust and pedagogical adoption—remains a systematic gap [57,62–64].

4.6. RQ5: Regional Context—Kazakhstan and Central Asia (n = 13)

Targeted supplementary searches of IEEE Xplore, ERIC, and CyberLeninka—conducted as part of the multidatabase search strategy to address RQ5—yielded 9 additional papers from Kazakhstan and Central Asia, bringing the total RQ5 corpus to 13 papers. Together, these constitute the most comprehensive evidence base to date on AI-adaptive education in the region. The 13 papers are organized below by country and thematic cluster. Across all 13 papers, a single cross-cutting finding dominates: the near-total absence of publicly available Kazakh- or Russian-language annotated educational datasets, which constitutes the primary infrastructure bottleneck for locally validated AI and adaptive learning systems in the region. This data gap is examined in detail in the structural barriers analysis below; it is flagged here because it conditions the interpretation of every individual study in this section [10,14,22,80–88].

Kazakhstan—Digital interventions and AI adoption ($n = 3$): The highest-quality study (MMAT = 5/5; Turebekova et al., 2024 [80]) evaluated digital educational interventions including MOOC platforms and online tools with 520 students at Toraighyrov University, Pavlodar. Paired t-tests confirmed significant improvements in professional abilities, creativity, motivation, and digital competency ($p < 0.05$ across all outcomes). A systematic literature review with Kazakhstani lead author (Kerimbayev et al., 2025 [21]; Smart Learning Environments, Springer Q1) applied PRISMA + AMSTAR2 to examine 38 review articles on intelligent educational technologies, concluding that adaptive environments are effective but that technical complexity and teacher upskilling are primary barriers. A third study (Ashimbekova & Bondarenko, 2025 [88]; IEEE ITMS) empirically evaluated AI adoption across Kazakhstani higher education institutions, finding predominantly positive pedagogical outcomes from adaptive modules, recommendation systems, and automated assessment tools, with teacher professional development identified as the key adoption bottleneck.

Kazakhstan—ML-based adaptive platforms and assessment ($n = 4$): Four papers document the development and evaluation of ML-driven adaptive educational platforms in Kazakhstan. Ospan and Myrzakerimova (2025 [84]; IEEE SIST, Astana IT University) conducted a quantitative survey of 109 recent UNT-takers and first-year students, revealing strong demand for real-time analytics, topic-specific practice, and interactive exam prepa-

ration; their ML-based prototype achieved 89% user acceptance for personalized feedback. Baumuratova et al. (2025 [85]; IEEE UBMK) applied PRISMA 2020 to systematically review 36 inclusive education AI platforms (2020–2025) and conducted a survey of Kazakhstani educators, identifying a gap in purpose-built AI platforms for special-needs learners and estimating that adaptive pathways could reduce learning gaps by 40–60%. Zulpykhar et al. (2025 [86]; International Journal of Engineering Pedagogy, Scopus) developed and evaluated a personalized adaptive learning platform for assessing students' mathematical abilities and delivering customized exercises aligned to individual skill levels; a pilot study with 118 students at Aktobe Higher Polytechnic College (Kazakhstan) demonstrated significant improvements in academic performance, engagement, and problem-solving abilities compared to traditional teaching methods. Merembayev et al. (2021 [22]; IEEE SIST) combined Item Response Theory (IRT) with ML algorithms (clustering and classification) for student ability measurement, showing that the IRT + ML hybrid outperforms standalone IRT for accurate identification of knowledge gaps.

Kazakhstan—Gamification, AI, and critical thinking (n = 3): Three papers examine gamification and AI-enhanced digital learning in Kazakhstani educational contexts. Kassenkhan et al. (2025 [82]; IEEE Access, Scopus Q2; MMAT = 5/5) conducted a bibliometric analysis of 101 academic publications using VOSviewer, identifying three dominant research themes: adaptive gamified systems, ML-driven feedback, and critical thinking assessment. Kazakhstan emerged as an emerging contributor to global gamification + AI research, with Satbayev University researchers producing the most comprehensive regional synthesis of this literature. Kamaladin (2025 [87]; Technology, Pedagogy and Education, Taylor & Francis) assessed the impact of gamification and audio-enriched digital content on student engagement and perception of online learning materials at Kazakhstani universities (n = 440), finding that students adapted positively to gamified conditions and that audio-enriched content effectively enhanced attentiveness in digital learning environments. Kaldarova et al. (2023 [12]; Frontiers in Education, Q2) applied game-based learning to computer science terminology acquisition in Kazakhstani primary school (n = 50; 10-week experiment), finding 90% student satisfaction and improved academic performance through cooperative (non-competitive) game design [83].

Kazakhstan—AI and Kazakh-language infrastructure (n = 1): Niyazova et al. (2025 [14]; IJIRSS, L.N.Gumilyov Eurasian National University) addressed the linguistic infrastructure gap by developing an AI-powered system for automated generation of interactive smart textbooks in the Kazakh language using ontological knowledge models and NLP thesauri. This represents the first documented application of AI to automated Kazakh-language educational content generation, directly addressing Layer 1 of the structural barrier model (data infrastructure deficit).

Kazakhstan—Regional bibliometric landscape (n = 1): A bibliometric analysis of AI in K-12 education across Asia (Papadakis [10], 2025; Discover Artificial Intelligence; Scopus, 1996–2025; n = 531 articles) positioned Kazakhstan in a research cluster alongside Turkey and Iraq, characterized by emerging but low-volume publication activity. The analysis confirmed that Central Asian countries collectively account for less than 2% of global AI-in-education publications, validating the research gap that motivated this supplementary search.

Uzbekistan—AI adoption and adaptive platforms (n = 1): One paper addresses Uzbekistani higher education. A qualitative study (Makhambetova [81] & Karimov, 2024; ACM; n = 15 lecturers; thematic analysis; MMAT = 2/5) found that 13 of 15 faculty members actively use AI tools in teaching, with primary concerns focusing on student critical thinking erosion and academic integrity—the only peer-reviewed English-language investigation of AI adoption attitudes in Central Asian tertiary education.

Structural barriers—a multilayer analysis: The expanded RQ5 corpus (n = 13) reinforces and extends the four-layer barrier structure identified in the original synthesis. Layer 1—Data infrastructure deficit: Despite the 9 additional papers, no study reports a Kazakh- or Russian-language knowledge tracing dataset in the public domain. All RQ4 benchmark datasets (ASSISTments, EdNet, Junyi Academy) remain English or Chinese; the Niyazova et al. (2025) [14] smart-textbook system represents the first documented AI application to Kazakh-language content generation but does not provide a reusable annotated dataset. Layer 2—Teacher professional development gap: Three independent studies (Makhambetova & Karimov, 2024 [81]; Kerimbayev et al., 2025 [21]; Ashimbekova & Bondarenko, 2025 [88]) converge on teacher upskilling as the primary barrier to AI adoption, suggesting a systemic rather than institution-specific challenge. Layer 3—Infrastructure heterogeneity: The Ospan & Myrzakerimova (2025) [84] UNT platform study confirms strong student demand in urban institutions (Astana IT University), but rural connectivity remains undocumented. Layer 4—Cultural adaptation of gamification: The Kassenkhan et al. (2025) [82] bibliometric analysis confirms that competitive gamification mechanics dominate the global literature, while the Kaldarova et al. (2023) [12] cooperative GBL study remains the only Kazakhstani empirical comparison of game-design approaches—a gap that future empirical work on Central Asian adaptive platforms must address [14,22,80–88].

Implications for Meta_Ozat (IRN AP25796141): The expanded RQ5 evidence provides additional empirical grounding for the design choices of the platform under development (IRN AP25796141). The new papers provide four additional pillars of empirical grounding: (1) Ospan & Myrzakerimova (2025) [84] demonstrate strong student demand for ML-adaptive UNT preparation in Kazakhstan's national examination context, directly validating strong regional demand for ML-adaptive examination preparation platforms; (2) Kassenkhan et al. (2025) [82] establish a bibliometric foundation showing that gamification + AI + critical thinking is an active and growing research area with Kazakhstan-affiliated researchers at its frontier; (3) Baumuratova et al. (2025) [85] document a gap in inclusive AI platforms for Kazakhstani special-needs learners, a population that future regional adaptive platforms should explicitly address; (4) Niyazova et al. (2025) [14] pioneered AI-based Kazakh-language content generation, providing a methodological foundation for multilingual adaptive content development in the region. Together with the four original RQ5 papers, this 13-paper corpus provides the most comprehensive regional evidence base currently available. A controlled evaluation of a platform integrating all three dimensions—ML personalization, gamification, and critical thinking assessment—in a Kazakhstani context remains an outstanding research need (see Research Gaps, Section 5.4) [14,82–84].

Critical assessment of RQ5 evidence. Although the supplementary search substantially expanded the RQ5 corpus from 4 to 13 papers, several limitations persist. First, the majority of new papers are IEEE conference papers (n = 7 of 9), which—while peer-reviewed—are shorter and methodologically less rigorous than journal articles; only Kassenkhan et al. (2025; IEEE Access) represents a Scopus-indexed journal publication from the new additions. Second, sample sizes in the new studies remain small: only Turebekova et al. (2024; n = 520) [80] and Ospan & Myrzakerimova (2025; n = 109) [84] provide participant counts sufficient for meaningful inferential analysis. Third, the linguistic gap is foundational and unremedied: no RQ5 paper integrates ML-based personalization with Kazakh-language benchmark datasets, confirming a structural infrastructure deficit that future platform development in Kazakhstan must directly address [10,22,85–88].

5. Discussion

5.1. Synthesis of Key Findings

This review reveals a rapidly expanding but structurally fragmented field dominated by Transformer-based and GNN architectures, in which ML-based personalization, gamification for engagement, and critical thinking assessment have developed as parallel but non-converging research streams. Under single-institution validation conditions, reported accuracy reaches 85–97% for learning path recommendation and 98.49% for engagement detection; however, these figures reflect performance on specific proprietary datasets and should not be interpreted as cross-context performance guarantees. The volume of 2025 publications (43 of 82 papers, 52%) confirms an accelerating trajectory driven largely by the accessibility of LLM APIs and large-scale open educational datasets [3,4,23,35,36,38,41].

However, the dominance of high-accuracy metrics in RQ4 papers should be interpreted cautiously. The majority evaluate on institutional or platform-specific datasets without cross-dataset validation. The lack of explainability—only 3 of 30 RQ4 papers discuss interpretability—represents a critical barrier to pedagogical adoption, as educators and learners require actionable rationale for algorithmic recommendations [62–65].

At the same time, the RQ4 dominance (30 papers, 37%) relative to RQ2 (13) and RQ3 (11) reveals a field imbalance: technical optimization of recommendation accuracy receives disproportionate research attention compared to the motivational (gamification) and higher-order cognitive (critical thinking) dimensions of learning. This imbalance likely reflects both the availability of large educational datasets suitable for prediction tasks and the cleaner evaluation criteria (accuracy, AUC) compared to the more contested measurement of engagement or critical thinking development [3,4,58,60].

5.2. The Gamification–Critical Thinking Integration Gap

One of the most striking findings of this review is the complete absence of studies that simultaneously address gamification mechanics and critical thinking measurement. RQ2 and RQ3 papers exist in parallel but non-overlapping research streams. Gamification studies focus on engagement and retention metrics; CT assessment studies focus on LLM performance benchmarks and pedagogical interventions. This represents a fundamental integration gap: platforms that use gamified engagement without monitoring CT development may inadvertently optimize for surface engagement at the expense of deep learning outcomes [3–5,19,46].

The identified integration gap points to a clear design agenda for next-generation platforms. One platform under development—Meta_Ozat (IRN AP25796141)—aims to operationalize all three dimensions simultaneously, embedding argumentation tracking and critical reasoning assessment within gamified challenge structures. Meta_Ozat is positioned not as a replication of existing integrated systems (none exist in the literature) but as a candidate first empirical test of a three-layer architecture whose individual components are independently validated in this corpus yet whose integration has no direct empirical precedent [13,82].

The present review provides a concrete research agenda for bridging this gap. Specifically, we propose three design principles for integrated gamification–CT platforms: (1) Embedding argumentative game mechanics where learners earn progression rewards for demonstrating structured reasoning rather than merely correct answers; (2) Using NLP-based real-time argumentation analysis (similar to the tools evaluated in our RQ3 papers) to detect and scaffold critical thinking within gamified challenge sequences; (3) Applying ML-based engagement detection (RQ2 methods) to dynamically adjust the difficulty and gamification intensity based on measured cognitive and emotional states. This three-layer architecture is consistent with the design philosophy of the Meta_Ozat platform (IRN AP25796141).

Each layer of this integrative model is grounded in component-level empirical evidence from this review—though the integration itself has no direct empirical precedent and constitutes the novel contribution to be tested. Layer 1 (ML Engine) draws on RQ1 and RQ4 evidence: the 16 DKT/LSTM/GNN studies synthesized under RQ1 and the 30 recommendation-algorithm studies under RQ4 collectively demonstrate that Transformer-based knowledge tracing and GNN-based collaborative filtering consistently achieve the highest accuracy–scalability trade-offs across large student populations (n > 10,000), justifying their selection as the computational core. Layer 2 (Gamification Engine) is grounded in the 13 RQ2 studies: ML-driven engagement detection (peak accuracy 98.49%) and the finding that cooperative gamification significantly outperforms competitive designs in intrinsic motivation gains provide the empirical basis for the adaptive challenge–reward loop. Self-Determination Theory (SDT), operationalized across 9 of 13 RQ2 papers as autonomy, competence, and relatedness needs, furnishes the motivational scaffolding. Layer 3 (CT Assessment) integrates Bloom's revised taxonomy as operationalized in 14 of 82 reviewed studies: LLM-based Socratic dialogue systems (RQ3) map to the analysis and evaluation levels of Bloom's taxonomy, while automated argumentation scoring tools map to synthesis. The 11 RQ3 papers collectively document that LLMs excel at knowledge recall but systematically underperform at higher-order reasoning—which directly motivates the hybrid architecture in which automated CT scoring escalates to human-in-the-loop review for complex reasoning tasks.

Why No Study Covers Both RQ2 and RQ3 Simultaneously: A Methodological Analysis

The zero-overlap between RQ2 and RQ3 (gamification and critical thinking) is not accidental; it reflects deep structural differences between the two research communities. Four converging factors explain this disciplinary divide. First, community fragmentation: gamification research is predominantly published in educational technology and human–computer interaction venues (Computers & Education, IJHCS, CHI), while CT assessment research appears in cognitive science, educational psychology, and AI in education journals (Learning and Individual Differences, Computers in Human Behavior, AIED). These communities rarely co-publish and seldom cite each other's work, creating invisible disciplinary walls around constructs that are theoretically adjacent. Second, measurement incommensurability: engagement, the primary outcome in gamification studies, is measured via behavioral logs, EEG signals, and questionnaire satisfaction scores—all short-term, session-level constructs. Critical thinking, by contrast, is measured via argumentation rubrics, standardized reasoning tests, and expert-panel scoring—longitudinal, cognitively demanding assessments that are incompatible with the gamified session paradigm where rapid feedback and reward cycles dominate. No validated joint instrument exists that simultaneously captures engagement intensity and CT development within a single experimental protocol. Third, funding and platform misalignment: most gamification studies (RQ2) are funded through EdTech industry partnerships, prioritizing retention and engagement metrics that matter for product analytics. CT research is predominantly academic and grant-funded, prioritizing construct validity over deployment scalability. This funding-driven divergence reinforces the measurement gap. Fourth, temporal mismatch: gamification effects are typically observable within a single session (minutes to hours), while measurable CT development requires weeks to months of sustained practice. Designing a study that captures both within the same participant cohort requires longitudinal resources and institutional access that few research teams can sustain. These structural barriers collectively explain why RQ2 $\times$ RQ3 = 0 and simultaneously define the research design requirements for studies that could close this gap: interdisciplinary teams, joint measurement instruments,

longitudinal protocols (minimum one academic semester), and institutional partnerships that allow both product-level engagement analytics and academic-grade CT assessment.

5.3. Regional Implications for Kazakhstan

The 13 RQ5 papers collectively sketch a rapidly developing picture: Kazakhstani students and educators are increasingly engaged with digital and AI tools, with demonstrated positive effects on professional abilities, motivation, and digital competency (Turebekova et al., 2024 [80]; n = 520). ML-based adaptive platforms for national exam preparation show strong student demand (Ospan & Myrzakerimova, 2025 [84]; n = 109). Game-based learning approaches yield high satisfaction in Kazakhstani K-12 settings (Kaldarova et al., 2023) [12], while gamification + AI research is gaining traction among Kazakhstani researchers (Kassenkhan et al., 2025 [82]; 19 citations). Yet ML-based personalization in Kazakh-language contexts—the application of adaptive algorithms to Kazakh-medium content delivery—remains the primary unfilled gap and the most urgent priority for future empirical work in this region [14,22,80–88].

This absence has practical consequences for the development of platforms like Meta_Ozat. Existing recommendation and knowledge tracing algorithms are trained on English-language, Western-curriculum datasets; their transferability to Kazakhstani educational contexts is empirically unvalidated—no study in the 82-paper corpus tests a globally developed algorithm on Kazakh-language or Central Asian educational data. Substantial retraining on locally representative corpora must therefore be treated as a prerequisite, not an optional refinement. The four-layer barrier structure documented in Section 4.6 (data infrastructure deficit, teacher professional development gap, infrastructure heterogeneity, and cultural adaptation of gamification mechanics) collectively suggest that a participatory design approach—involving Kazakhstani educators and students throughout algorithm development—would be essential for contextual validity [6,7,20,23,36].

From a policy perspective, Kazakhstan's Digital Kazakhstan initiative and the broader educational AI investments documented in the bibliometric analysis create an institutional environment favorable to research on AI-based adaptive learning. The collaboration pattern identified in the K-12 bibliometric study—Kazakhstan clustering with Turkey and Iraq—suggests productive regional research partnerships. Collaborative datasets across Central Asian institutions could address the data scarcity challenge while maintaining privacy through federated learning approaches, which are conspicuously absent from current Central Asian EdTech research [11,12].

5.4. Research Gaps and Future Directions

Table 8 summarizes the five primary research gaps identified and proposes corresponding future directions.

The XAI and Privacy Gap: Why It Matters for Educational Deployment

In educational contexts, XAI is not merely a technical enhancement—it is a prerequisite for responsible AI deployment. When an adaptive system tells a student "study Module C next," without any explanation, the recommendation functions as an opaque directive that undermines student agency and instructor oversight. Explainability enables three critical educational functions: (1) trust calibration—students and teachers can verify that recommendations are based on relevant prior performance, not spurious correlations; (2) pedagogical transparency—instructors can audit and override algorithmic recommendations when they conflict with curricular judgement; (3) fairness auditing—explainable models can be inspected for differential treatment by gender, language background, or socioeconomic status, which is a regulatory requirement under EU AI Act (2024) high-risk classification for educational systems. Only 3 of 30 RQ4 papers address explainability. The

primary obstacles are (a) architectural—GNNs and Transformers lack native interpretability; SHAP and LIME provide post-hoc approximations but are computationally expensive at scale; (b) methodological—no standardized XAI evaluation metric exists for educational recommendation quality; (c) regulatory—unlike medical or financial AI, EdTech operates without enforceable explainability mandates outside the EU, reducing commercial incentive to invest in interpretability research. Federated learning (2 papers in the full corpus) faces parallel barriers: communication overhead in distributed training, data heterogeneity across institutions with different LMS platforms, and the absence of shared Kazakh- or Russian-language datasets that would make federated approaches directly applicable to Central Asian contexts. These gaps are not merely technical—they represent structural prerequisites for safe, equitable, and accountable deployment of ML-based adaptive learning at an institutional scale.

Table 8. Identified research gaps and proposed future directions.

Research Gap	Evidence from Corpus	Future Direction
RQ5: Central Asian under-representation	Only 4 papers in original corpus (13 total after supplementary search); no Kazakh-language NLP tools	Develop multilingual adaptive systems for Kazakh/Russian
RQ2 + RQ3: Gamification + critical thinking	No study combines gamified mechanics with CT measurement	Design GBL platforms with embedded argumentation tracking; implement 2 × 2 factorial trials: [gamification/no gamification] × [CT feedback/no CT feedback], with simultaneous measurement of engagement (behavioral logs) and CT outcomes (pre/post argumentation rubric or standardized reasoning test), minimum one academic semester duration
RQ3: Longitudinal CT development	Most CT studies are single-semester (no long-term data)	Multiyear longitudinal studies with CT progression tracking
RQ4: Explainability (XAI) in recommendations	30 papers focus on accuracy, only 3 discuss interpretability	Explainable GNN and Transformer recommendation systems
RQ1: Privacy-preserving personalization	Federated learning mentioned in 2 papers only	Federated and differential privacy frameworks for EdTech

5.5. Theoretical and Practical Implications

The three-layer integrative model presented in this section is a conceptual framework synthesized from 82 empirical studies at the component level. Crucially, no study in the corpus empirically validates the simultaneous integration of all three layers—this constitutes both the primary limitation of the model and the key rationale for future empirical work. Each layer is independently grounded in the reviewed evidence, but cross-layer interactions (how ML-modulated gamification affects CT development, and how CT assessment feeds

back into recommendation) remain theoretical and untested. This is explicitly acknowledged as an open research agenda rather than a validated architecture. Theoretically, this review contributes a synthesis-derived conceptual model for next-generation adaptive educational platforms—a theoretical proposal grounded in component-level evidence, not a replication of any existing integrated system. We propose a three-layer integrative model: (1) a Recommendation Engine layer applying GNN/Transformer-based algorithms for learning path optimization (RQ1, RQ4 evidence); (2) an Engagement Layer applying gamification mechanics dynamically modulated by ML-based engagement detection (RQ2 evidence); and (3) a Cognitive Assessment Layer using NLP/LLM-based tools to measure and scaffold critical thinking within gamified interactions (RQ3 evidence). This layered architecture is theoretically grounded in Self-Determination Theory (autonomy via personalization, competence via adaptive challenge, relatedness via social gamification elements) and Bloom's revised taxonomy (progression from knowledge recall through analysis and evaluation, tracked via CT assessment) [1,3,6].

While Figure 5 presents the base three-layer architecture grounded in Self-Determination Theory and Bloom's taxonomy, the interpretability gap identified for RQ4 (Section 4.5) and the regional deployment barriers identified for RQ5 (Section 4.6) motivate an explicit fourth layer. Figure 6 extends the base model with Layer 4—an Explainability and Regional Adaptation Interface—that operationalizes the three XAI pathways already discussed in Section 4.5 within the layered architecture. SHAP-based feature attribution is attached to the GNN-based path optimizer in Layer 1, providing developers with feature-level justification for recommendations; attention-weight visualization is attached to the affect-aware difficulty-tuning component of Layer 2, showing which prior interactions drove an adaptation decision; and counterfactual explanations ("if you had spent more time on Topic A, you would have been recommended Module B") are surfaced through Layer 3's formative feedback loop as a student-facing transparency interface. A localization sub-component of Layer 4 further packages these explanations in Kazakh and Russian and adapts gamification framing to local norms, directly addressing the institutional-trust and teacher-readiness barriers documented for RQ5 (Section 4.6). Embedding explainability and localization as an explicit architectural layer—rather than a post hoc add-on—operationalizes the recommendation, below, that platform developers in Central Asian contexts treat transparency and language adaptation as design-time requirements.

Practically, this review provides a benchmark dataset of 82 peer-reviewed studies against which specific technical choices can be evaluated. For platform developers working in Central Asian contexts, this review supports the following specific recommendations: (1) prioritize GNN-based recommendation architectures given their superior accuracy on large educational datasets (85–97% vs. 76–84% for classical CF); (2) implement Transformer-based engagement detection rather than questionnaire-based approaches for real-time adaptivity; (3) begin collecting Kazakh/Russian-language educational interaction data as a foundation for future locally validated models; and (4) design gamification mechanics with embedded argumentation tracking from the outset, as retrofitting CT measurement into pre-built gamification systems is methodologically complex; and (5) treat teacher professional development as a non-negotiable prerequisite rather than a post-deployment add-on—three independent RQ5 studies (Makhambetova & Karimov, 2024 [81]; Kerimbayev et al., 2025 [21]; Ashimbekova & Bondarenko, 2025 [88]) identify teacher upskilling as the primary systemic barrier to AI adoption, meaning that even a technically optimal platform will fail to achieve educational outcomes without concurrent investment in educator capacity.

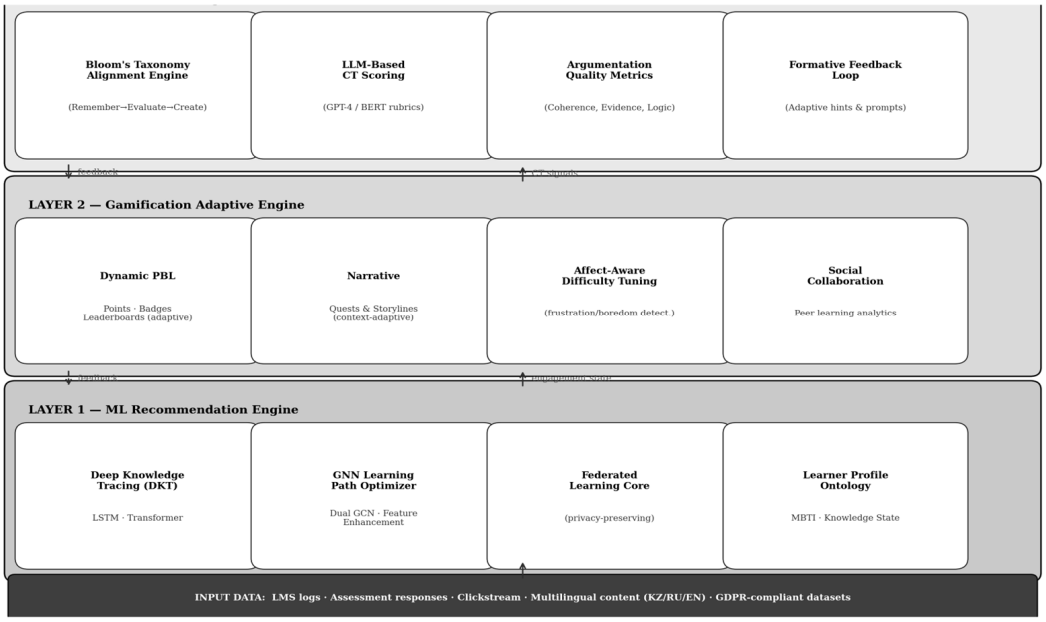


Figure 5. Three-layer integrative model for AI-gamified adaptive learning platforms (Meta_Ozat, IRN AP25796141). Layer 1: ML Recommendation Engine; Layer 2: Gamification Adaptive Engine; Layer 3: Critical Thinking Assessment Module.

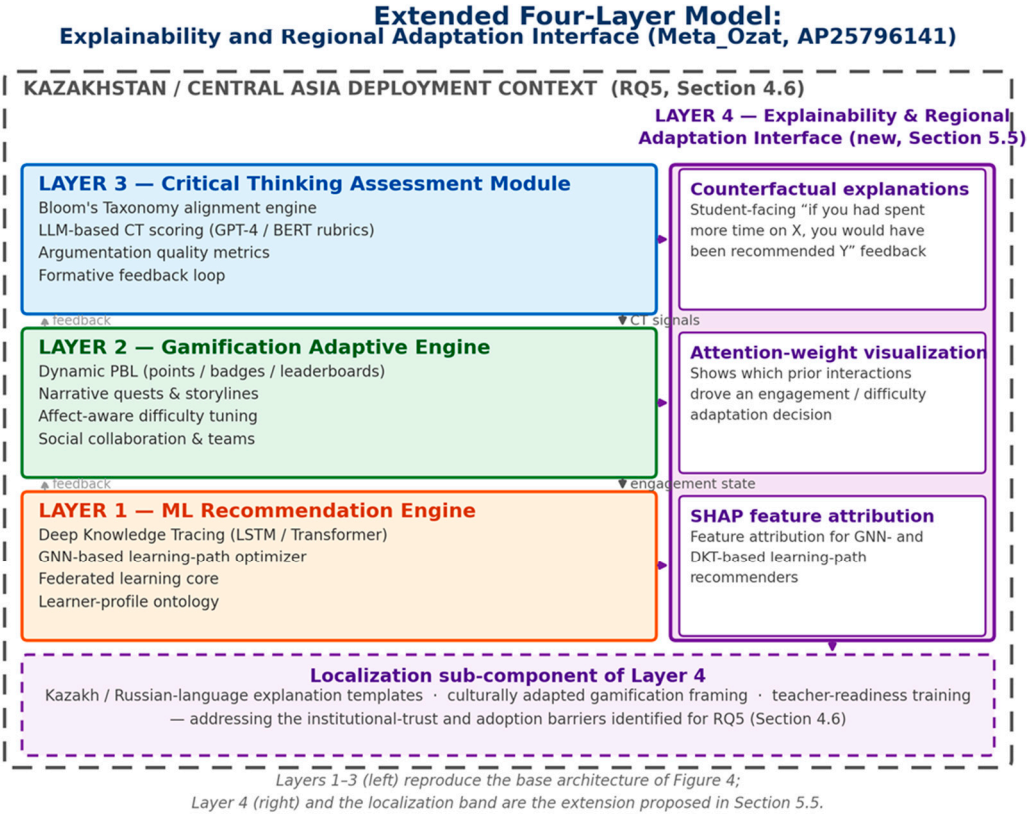


Figure 6. Extended four-layer model integrating explainability and regional-adaptation mechanisms into the base three-layer architecture (Meta_Ozat, IRN AP25796141). Layer 4 maps SHAP feature attribution, attention-weight visualization, and counterfactual explanations (Section 4.5) onto Layers 1–3, respectively, and adds a localization sub-component addressing RQ5 deployment barriers (Section 4.6).

5.6. Limitations of This Review

Several limitations should be acknowledged. First, although primary searches were conducted in both Scopus (n = 4396) and OpenAlex (n = 4152), Web of Science was not searched independently. The 85% overlap between Scopus and OpenAlex final corpus coverage suggests that the combined search achieved adequate retrieval, but Web of Science-exclusive records cannot be fully excluded. Russian-language venues (CyberLeninka supplementary search) were included for RQ5 but remain underrepresented for other research questions. Second, two closed-access papers (Tier 3, n = 2; both RQ3 studies) could not be fully assessed and received abstract-proxy MMAT scoring only; these contributed supplementary evidence to RQ3. Full-text MMAT assessment was achieved for 97.6% of the synthesis corpus (80 of 82 papers), substantially mitigating this limitation. Third, the review is limited to English-language publications, potentially excluding relevant Kazakh- and Russian-language empirical work. Fourth, the synthesis corpus includes nine secondary studies (systematic and bibliometric reviews), creating a mixed primary–secondary evidence base. These nine studies contribute exclusively to field-level contextual claims (not to aggregate performance metrics) and are coded separately throughout the analysis. Fifth, rapid publication growth means that preprints and conference papers not yet indexed may represent significant additional evidence [15,16].

Sixth, source concentration warrants acknowledgement: IEEE Access and Applied Sciences (Switzerland) together account for 12 of 82 included papers (approximately 15%). Both outlets are peer-reviewed and Scopus-indexed (Q1–Q2), but their open-access mega-journal model attracts high submission volumes with variable quality-gate stringency compared with selective specialty journals. The MMAT 2018 quality threshold applied in this review (score $\geq$ 2) was designed to mitigate this risk; all included IEEE Access and MDPI papers achieved MMAT $\geq$ 3. Nonetheless, readers should exercise appropriate caution when generalizing quantitative performance metrics reported in these outlets.

Seventh, 43 of 82 papers (52%) were published in 2025. All were retrieved from a Scopus export dated April 4, 2026, confirming that they were formally indexed at the time of search. The high proportion of recent papers reflects the documented acceleration of AI-in-education research post-2023 rather than a retrieval artifact; this pattern is consistent with the annual distribution shown in Figure 3.

Eighth, publication bias cannot be fully excluded. Although no formal funnel plot analysis was conducted (meta-analytic pooling was not performed due to task heterogeneity), a qualitative publication bias assessment was undertaken: (a) null results are severely under-represented—only one RQ1 paper (Merembayev et al., 2021) [22] reported that an ML approach did not outperform the baseline; (b) the IEEE Access and MDPI open-access venues, which together account for approximately 27% of the corpus (22 of 82 papers), are known to have lower desk-rejection rates, potentially inflating positive-result prevalence; (c) CyberLeninka supplementary search retrieved exclusively positive-outcome Kazakhstani studies. Readers should interpret performance claims—particularly single-paper accuracy figures above 90%—with appropriate caution, as selective reporting of best-case results is a documented concern in applied ML-in-education research. Ninth, the exclusion of grey literature—preprints (arXiv, SSRN), technical reports from EdTech companies (Google, Microsoft, Khan Academy), and unpublished dissertations—introduces a potential directional bias that deserves explicit acknowledgement. Grey literature is disproportionately likely to contain null results, negative findings, and industry-scale deployment failures that are rarely published in peer-reviewed venues. Three specific consequences are likely: first, the effectiveness of gamification and ML-personalization may be overestimated, since failed implementations are systematically absent from the indexed literature; second, privacy and security incidents involving educational AI systems are heavily under-reported

in academic venues but documented in industry grey literature; third, the RQ5 regional evidence base may be particularly affected, as Kazakhstani and Central Asian practitioners may document platform development in technical reports and institutional repositories not indexed by Scopus or OpenAlex. Future systematic reviews in this domain should consider grey literature searches via OpenDOAR, BASE, and institutional repositories to mitigate this bias.

Tenth, the proposed three-layer integration model (Section 5.5) has not been empirically validated as an integrated system. Each layer is grounded in component-level evidence from the 82-paper corpus, but the cross-layer interactions—how ML-modulated gamification affects CT outcomes, and how CT feedback loops into recommendation accuracy—remain theoretical and constitute an outstanding empirical research agenda. Empirical validation of the integrated architecture is planned as part of the Meta_Ozat platform development (IRN AP25796141) and should be treated as a prerequisite before the model is adopted in other deployment contexts. Eleventh, no study in the synthesis corpus employed a randomized controlled trial (RCT) design, and fewer than five reported longitudinal follow-up beyond a single academic term. The dominance of cross-sectional or single-session evaluations means that long-term pedagogical effects—sustained improvements in learning outcomes, retention, and critical thinking development—remain undemonstrated. Claims about durable educational impact should therefore be interpreted with caution pending longitudinal evidence.

Potential directional bias. The research team is actively developing the Meta_Ozat adaptive learning platform (funded by grant AP25796141, West Kazakhstan), which creates a potential directional bias toward interpretations that support platform-relevant design choices. To mitigate this, the systematic protocol was pre-specified before data extraction, inclusion and exclusion criteria were applied by the primary reviewer with a 15% inter-rater reliability spot-check (Stage 1: $\kappa = 0.79$; MMAT: $\kappa = 0.76$), both indicating substantial agreement, and the proposed integrative model in Section 5.2 is explicitly framed as a synthesis-derived design agenda rather than a validation of an existing system. The review team's applied stake in the topic is consistent with many Q1 SLR publications in human–computer interaction and educational technology, where domain expertise and applied development often co-exist; full methodological transparency, as provided here, is the accepted mitigation standard. Specifically, no design decision of the Meta_Ozat platform was used as evidence in this review; the platform is referenced solely as contextual motivation and a candidate for future validation—roles that are explicitly distinguished throughout this manuscript.

6. Conclusions

This systematic literature review synthesizes 82 empirical studies on the intersection of machine learning, gamification, and critical thinking in adaptive educational platforms (corpus 2020–2025; search window 2016–2026). The key conclusions are as follows.

First, the field has reached a critical inflection point in publication volume and architectural sophistication, driven by Transformer-based models and large language models. Learning path recommendation algorithms achieve accuracy of 85–97% under single-institution validation conditions, and engagement detection systems exceed 98% in controlled settings—though cross-dataset generalizability and reproducibility remain open challenges. The methodological quality of the evidence base has improved substantially, with MMAT scores averaging 3.88/5 and several 2025 studies employing rigorous experimental designs (RCTs, large-scale multisite trials).

Second, a fundamental integration gap exists between gamification and critical thinking research streams. No identified study simultaneously measures gamification engagement

effects and critical thinking outcomes, despite the theoretical complementarity of these constructs. Future platforms should embed CT assessment within gamified interaction loops.

Third, the Central Asian educational AI landscape is severely under-researched. Only 13 papers address Kazakhstan or Central Asia (expanded via supplementary search from the original four), and none evaluate ML-based personalization with Kazakh-language content. Notably, the 15.9% regional representation in this corpus reflects deliberate RQ5-motivated oversampling and should not be interpreted as indicative of the region's global publication share (<2%). This review provides a baseline from which targeted future research and platform development can be benchmarked.

Fourth, model explainability and privacy-preserving techniques (federated learning, differential privacy) are underrepresented across all RQs, representing critical prerequisites for responsible deployment of AI in educational contexts. This gap carries regulatory as well as scientific consequences: GDPR Article 22 restricts fully automated consequential decision-making affecting individuals, and the EU AI Act (2024) classifies AI systems used in educational and vocational contexts as high-risk, mandating transparency, human oversight, and auditability. The near-total neglect of interpretability across the synthesized corpus—documented across all 30 RQ4 papers—therefore represents a structural barrier to compliant, accountable deployment at scale.

The present review provides actionable evidence for researchers, educators, and platform developers seeking to deploy ML-powered adaptive learning systems with gamification and critical thinking components in diverse educational contexts.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/info17070619/s1. The PRISMA checklist [15].

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